Linear Algebra Done Taiwan

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Contents

0.1 Preface

Done Taiwan Series is a series of books that aims to help students prepare for the entrance examinations for top mathematics master's programs in Taiwan. The series currently consists of two books: *Linear Algebra Done Taiwan* and *Advanced Calculus Done Taiwan*. Each includes solutions and corrections to the exams administered by National Taiwan University over the past 10 years.

The one you are currently reading is *Linear Algebra Done Taiwan*. The two books differ significantly in structure and style. The first four chapters of this book develop the theory of linear algebra the most elegant way possible. As their titles suggest,these chapters are arranged roughly in order of importance. Personally, I can promise you that if you understand all the materials in the first four chapters, you will score near perfect on the subject matter. Throughout this book, $\mathbb{F}$ always denotes a field, and V always denotes a vector space over $\mathbb{F}$.

In this book, you will constantly see brown text like this:

Click me to go to the last page of the book

They are to make cross-references easier to follow. For example, you may read something like "By primary decomposition theorem, we know...", and clicking primary decomposition theorem would lead you to a statement and possibly a proof of primary decomposition theorem.

Just like all mathematical texts, this book may contain some minor errors. If you believe you see an error or are confused by my writing, feel free to contact me. My email address is:

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Chapter 1

Fundamentals

1.1 Quick recap

Let V be a vector space over $\mathbb{F}$, and let S be a subset of V. We say

(a) S is **linearly independent** if to write 0 as finite linear combination of S, the coefficients must all be zero.

(b) S **spans** V if all $v \in V$ can be written as finite linear combination of S.

We say S is a **basis** if any of the following equivalent conditions hold true

(a) S is a maximum linearly independent subset of V.

(b) S is linearly independent and spans V.

(c) All elements of V can be uniquely written as some finite linear combination of S.

Suppose V has a basis of cardinality n. Using **Gauss elimination**, one see that all linear independent subset of V must have cardinality not greater than n, and all basis must have cardinality n. Therefore, for vector space with some basis of finite cardinality, it make sense to refer to its **dimension**. Let $\dim(V) = n$. Each linearly independent subset S of V of cardinality n must be a basis, otherwise we may construct a linearly independent set of cardinality greater than n. By an algorithm, one see that

(a) If S spans V and has cardinality greater than n, then there exists a subset of S forming a basis of V.

(b) If B is a basis of V and S is linearly independent, then there exists some subset U of B such that $S \cup U$ is a basis of V.

Let W be another vector space over $\mathbb{F}$, and let $F \in \operatorname{Hom}(V, W)$. It is clear that both image and kernel of F forms a vector space. Therefore, it make sense to define the **rank** of F by

$$\operatorname{rank}(F) \triangleq \dim(\operatorname{Im} F).$$

Clearly, for all basis B of V, we have

$$|X| \leq \dim(\operatorname{Ker} F), \quad \text{where } X \triangleq \{v \in B : F(v) = 0\}$$

Moreover, letting $Y \triangleq \{v \in B : F(v) \neq 0\}$, because $X \cup Y$ forms a basis, we also have

$$\operatorname{span}\{F(v) \in W : v \in Y\} = \operatorname{Im}(F)$$

which implies

$$\operatorname{rank}(F) \leq |Y|$$

Selecting some subset S of Y that makes $\{F(v) \in W : v \in S\}$ a basis of $\operatorname{Im}(F)$, and replacing $v \in Y - S$ with $v - \sum c_i s_i$ so that $F(v) = 0$, we now see the new X forms a basis for $\operatorname{Ker} F$, and the new Y make $\{F(v) \in W : v \in Y\}$ a basis for W. This implies the **Rank-Nullity Theorem**:

$$\dim(V) = |X| + |Y| = \dim(\operatorname{Ker} F) + \operatorname{rank}(F)$$

We use $V^\vee$ to denote the **dual space** of V. Let $F \in \operatorname{Hom}(V, W)$. Its **dual map** $F^\vee \in \operatorname{Hom}(W^\vee, V^\vee)$ is defined by

$$F^\vee(\varphi) \triangleq \varphi \circ F$$

If V is finite-dimensional with basis $\{v_1, \dots, v_n\}$, then its **dual basis** $\{\varphi_1, \dots, \varphi_n\}$ is defined by

$$\varphi_i(v_j) \triangleq \delta^i_j$$

and there exists a natural isomorphism between V and its double dual $(V^\vee)^\vee$ by identifying $v \in (V^\vee)^\vee$ as

$$v(\varphi) \triangleq \varphi(v).$$

Let $\{w_1, \dots, w_m\}$ be a basis of W with dual basis $\{\xi_i, \dots, \xi_m\}$. It is clear that under these two basis, the matrices representation of F and $F^\vee$ are *transpose* of each other. This together with the observation that $\operatorname{rank}(F) + \dim(\ker F^\vee) = \dim(W)$ for finite dimensional V and W, give us a quick proof that the dimensions of column space and row space of a fixed matrix are always equal. Now, given some n-by-n square matrix A, its **determinant** is by definition

$$\det A \triangleq \sum_{\sigma \in S_n} \Big(\operatorname{sgn} \sigma \prod_{k=1}^{n} A_{\sigma(k),k} \Big)$$

If one identify $M_n(\mathbb{F})$ with $(\mathbb{F}^n)^n$, determinant can be equivalently defined to be the unique alternating multilinear map from $(\mathbb{F}^n)^n$ to $\mathbb{F}$ such that

$$\det(I) = 1$$

By an algorithm, one see that, for each alternating multilinear map $F : (\mathbb{F}^n)^n \to \mathbb{F}$, we have

$$F(B) = (\det B) \cdot F(I) \text{ for all } B.$$

Therefore, if we observe for each $A \in M_n(\mathbb{F})$ that the map $B \mapsto \det AB$ is indeed alternating multilinear, then we immediately have the celebrated result $\det AB = \det A \det B$ as a corollary. This multiplicative property of determinant allow us to well define determinant for each linear endomorphism F of some finite-dimensional vector space by

$$\det F \triangleq \det([F])$$

where $[F]$ is the matrix representation of F with respect to some basis. Moreover, because the inverse of a linear map is linear if exists, we see that for linear map F, we have

$$F \text{ is invertible} \iff \det F \neq 0$$

and have

$$A \text{ is invertible} \iff \det A \neq 0$$

Given some $F \in \text{End}(V)$, we say $v \in V$ is an **eigenvector** with respect to the **eigenvalue** $\lambda \in \mathbb{F}$ if $v \neq 0$ and $F(v) = \lambda v$. If there exists some basis of V consisting eigenvectors of F, we say F is **diagonalizable**. Let A be some square matrix, if there exists some invertible square matrix P such that PAP^{-1} diagonal, we also say A is **diagonalizable**. Immediately, we see that if V is finite dimensional, then with respect to any basis of V

$$[F] \text{ is diagonalizable} \iff F \text{ is diagonalizable}$$

Let V be finite dimensional. We define the **characteristic polynomial** of F and A by

$$\det(tI - F) \text{ and } \det(tI - A)$$

Obviously, λ is an eigenvalue of F if and only if λ is a root of the characteristic polynomial of F.

1.2 Two forms

Abstract

In this section, V is a finite dimensional $\mathbb{F}$-vector space, and $T \in \mathrm{End}(V)$.

Theorem 1.2.1. (Structure theorem for finitely generated modules over PID) Let M be a finitely generated module over a PID R. Then there exists some nonnegative integer q and some tuple of proper nonzero ideals (f_i) of R that admits an R-module isomorphism:

$$M \cong R/(f_1) \oplus \cdots \oplus R/(f_n) \oplus R^q$$

Moreover, we know that

(i) Such q is unique, called the **rank** of M.

(ii) If we add the hypothesis of (f_i) being primary, then the tuple (f_i) is unique up to switching of order. Such isomorphism is then called the **primary decomposition of** M.

(iii) If we add the hypothesis $(f_i) \supseteq (f_{i+1})$, then the tuple (f_i) is unique. Such isomorphism is then called the **invariant factor decomposition of** M.

Proof. See Wikipedia. ■

As we promised in the preface, we will use the structure theorem for finitely generated modules over PID mentioned above to construct variant forms of endomorphisms of finite dimensional vector spaces. However, before such we need to check that the hypothesis are indeed satisfied:

Theorem 1.2.2. (What we need) We can make V a finitely generated $\mathbb{F}[x]$-module by defining the scalar multiplication by

$$x \cdot v \triangleq T(v)$$

Moreover, $\mathbb{F}[x]$ is a PID, and we have the followings:

(i) If $(f_i) \supseteq (f_{i+1})$ are two proper nonzero ideals of $\mathbb{F}[x]$, then f_i divides f_{i+1}.

(ii) If (f) is a proper nonzero primary ideal of R, then $f = g^n$ for some irreducible $g \in \mathbb{F}[x]$ and $n \in \mathbb{N}$.

Proof. Left to reader. ■

From now on in this section, we denote $R \triangleq \mathbb{F}[x]$. We first construct the **Frobenius normal form** of T. Because V is a finitely generated R-module, we may apply the structure theorem for finitely generated modules over PID to gain its invariant factor decomposition, a R-module isomorphism:

$$V \cong R/(f_1) \oplus \cdots \oplus R/(f_n) \oplus R^q$$

in which f_i divides f_{i+1}.

Theorem 1.2.3. (What we need) Every R-module is naturally a $\mathbb{F}$-vector space. Let $h : M \to N$ be a R-module homomorphism. If we consider M and N natural $\mathbb{F}$-vector spaces, then h is a $\mathbb{F}$-linear transformation.

Proof. Left to reader. ■

Now, since R as a natural $\mathbb{F}$-vector space is infinite dimensional and V is finite dimensional, we see that we must have $q = 0$. Therefore, we actually have

$$V \cong R/(f_1) \oplus \cdots \oplus R/(f_n)$$

Using this isomorphism and the fact that f_i divides f_{i+1}, one can check that the minimal polynomial of T is f_n.

Theorem 1.2.4. (Determinant of companion matrix) Let $f \in \mathbb{F}[x]$ be a monic polynomial:

$$f \triangleq x^d + c_d x^{d-1} + \cdots + c_1$$

By the **companion matrix** of f, we mean a matrix $A \in M_d(\mathbb{F})$ of the form

$$A_{i,j} = \begin{cases} -c_i & \text{if } j = d \\ 1 & \text{if } j = i - 1 \\ 0 & \text{if otherwise} \end{cases}$$

The characteristic polynomial of A is f.

Proof. This can be proved via induction, which we leave to reader. ■

Theorem 1.2.5. (The existence of Frobenius normal form of T) The $\mathbb{F}$-vector subspaces $R/(f_i)$ of V are all invariant under T. Fix i, and let f_i be monic with degree d. Then $\{1, \dots, x^{d-1}\} \subseteq R/(f_i)$ form a basis. Moreover, the matrix representation of the restriction of T onto the subspace $R/(f_i)$ with respect to this basis is the companion matrix of f_i. Therefore, the characteristic polynomial of T is $\prod_{i=1}^n f_i$.

Proof. Left to reader. ■

Theorem 1.2.6. (Primary decomposition theorem) Let p be the minimal polynomial of T, and write

$$p = p_1^{r_1} \cdots p_k^{r_k}$$

with p_i monic irreducible. Then V is the internal direct sum of the kernel of $p_i^{r_i}(T)$:

$$V \cong \ker(p_1^{r_1}(T)) \oplus \cdots \oplus \ker(p_k^{r_k}(T)) \tag{1.1}$$

Moreover, for each $i \in \{1, \ldots, k\}$, if we let $E_i : V \to \ker(p_i^{r_i}(T))$ be the projection along the rest of the direct sum, then there exists a polynomial q_i that makes $q_i(T) = E_i$. Because of such, for each T-invariant subspace W of V, we have

$$W = (W \cap \ker(p_1^{r_1}(T))) \oplus \cdots \oplus (W \cap \ker(p_k^{r_k}(T)))$$

Proof. Apply structure theorem for finitely generated modules over PID to get the primary decomposition form of the R-module V. That is, we get a set $\{p_1, \ldots, p_k\}$ of irreducible polynomials, for each $i \in \{1, \ldots, k\}$ a tuple $(m_{i,1}, \ldots, m_{i,n_i})$ of natural numbers, and an R-module isomorphism:

$$V \cong \bigoplus_{i=1}^{k} \bigoplus_{j=1}^{n_i} R \big/ (p_i^{m_{i,j}})$$

Equation 1.1 then follows from checking that

(i) The minimal polynomial of T is

$$\prod_{i=1}^{k} p_i^{r_i}, \qquad \text{where } r_i = \max\{m_{i,1}, \ldots, m_{i,n_i}\}$$

(ii) We have

$$\bigoplus_{j=1}^{n_i} R \big/ (p_i^{m_{i,j}}) = \ker(p_i^{r_i}(T))$$

Lastly, because p_i are monic irreducible, by Chinese remainder theorem there exists a polynomial $q_i \in \mathbb{F}[x]$ that makes:

$$q_i \equiv 1 \pmod{p_i^{r_i}} \quad \text{and} \quad q_i \equiv 0 \pmod{p_j^{r_j}}, \quad \text{for all } j \neq i$$

Such q_i then clearly satisfies $q_i(T) = E_i$. ■

From now on in this section, suppose $\mathbb{F}$ is algebraically closed. We close this section by constructing the **Jordan form** of T and prove the existence and uniqueness for **Jordan-Chevalley decomposition**. As in proof of primary decomposition theorem, let p be the minimal polynomial of T, and write

$$p = p_1^{r_1} \cdots p_k^{r_k}$$

with p_i monic irreducible. Because $\mathbb{F}$ is algebraically closed, we know p_i are linear. Therefore, we have a set $\{\lambda_1, \dots, \lambda_k\}$ of elements in $\mathbb{F}$ that makes

$$p_i = x - \lambda_i, \quad \text{for all } i$$

Theorem 1.2.7. (The existence of Jordan form of T) The $\mathbb{F}$-vector subspaces $R\diagup(p_i^{m_{i,j}})$ are all invariant under T. Fix i and $m_{i,j}$. Then

$$\left\{p_i^{m_{i,j}-1}, \dots, p_i, 1\right\} \subseteq R\diagup(p_i^{m_{i,j}})$$

forms a basis for $R\diagup(p_i^{m_{i,j}})$ such that the matrix representation of the restriction of T forms the Jordan block:

$$\begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix}$$

Proof. Left to reader.[1] ■

By **Jordan-Chevalley decomposition** of T, we mean a nilpotent $N \in \text{End}(V)$ and a diagonalizable $D \in \text{End}(V)$ that makes

(i) $T = D + N$.

(ii) D and N commute.

Corollary 1.2.8. (Existence and Uniqueness of Jordan Chevalley decomposition) The Jordan-Chevalley decomposition of T exists uniquely, and there exists some polynomial $p \in \mathbb{F}[x]$ that makes $D = p(T)$ and $p(0) = 0$.

Proof. The existence of Jordan-Chevalley decomposition of T follows from the existence of Jordan form of T. Before we prove the uniqueness, we first prove the existence of p. This follows from Chinese remainder theorem, in which we get a $p \in \mathbb{F}[x]$ that makes

$$p \equiv \lambda_i \pmod{(x - \lambda_i)^{r_i}}, \quad \text{for all } i$$

[1] If you think about it, the uniqueness of Jordan form follows from the uniqueness in the structure theorem for finitely generated modules over PID. The reason is that if you are given the Jordan form of T, you can get the primary decomposition of the R-module V.

Such p clearly makes $D = p(T)$. Now, if 0 is an eigenvalue, then we have $p \equiv 0 \pmod{x^{r_i}}$ for some i, so we have $p(0) = 0$. If 0 is not an eigenvalue, then the minimal polynomial of T has nonzero constant term, so adding appropriate amount of the minimal polynomial of T to p will make $p(0) = 0$. The existence of p implies that D and N commute, and the uniqueness of Jordan-Chevalley decomposition of T now follows from the fact that commuting diagonalizable linear operators are simultaneously diagonalizable[2]. ■

[2] Prove this via induction on dimension

1.3 Spectral theorem

Abstract

In this section, $\mathbb{F}$ is always a subfield of $\mathbb{C}$, and V is a $\mathbb{F}$-vector space.

By a **norm** on V, we mean some **positive-definite** functional $\|\cdot\|: V \to \mathbb{R}$ that satisfies **absolute homogeneity** and **triangle inequality**. In this context, by $\|\cdot\|$ being positive-definite, we mean

$$\|v\| = 0 \implies v = 0$$

Observing that

$$\|0\| = \|0 + v\| \leq \|0\| + \|v\|, \quad \text{for all } v \in V$$

we see norm must also be nonnegative. By an **inner product** on V, we mean some **positive-definite** map $\langle \cdot, \cdot \rangle : V^2 \to \mathbb{F}$ that is linear in the first argument and satisfies **conjugate symmetry**:

$$\langle v, w \rangle = \overline{\langle w, v \rangle} \quad \text{for all } v, w \in V$$

In this context, by $\langle \cdot, \cdot \rangle : V^2 \to \mathbb{F}$ being positive-definite, we mean

$$\langle v, v \rangle > 0 \quad \text{for all } v \neq 0$$

With an inner product equipped on V, we may discuss the **orthogonality** of vectors. Given a countable set $\{v_1, v_2, v_3, \dots\}$ of non-zero vectors, we use the term **Gram-Schmidt process** to refer to the process of defining

$$e_n \triangleq v_n - \frac{\langle v_n, e_{n-1} \rangle}{\|e_{n-1}\|^2} e_{n-1} - \cdots - \frac{\langle v_n, e_1 \rangle}{\|e_1\|^2} e_1$$

so that $\{e_1, e_2, e_3, \dots\}$ become orthogonal while

$$\operatorname{span}(v_1, \dots, v_n) = \operatorname{span}(e_1, \dots, e_n) \quad \text{for all } n.$$

Although implicit, Gram-Schmidt process may be considered one of the most important notions in inner product space. For example, the Gram-Schmidt process, together with **Pythagorean Theorem**, can be used to prove the **Cauchy-Schwarz Inequality**, which in term shows that the functional $f : V \to \mathbb{R}$ induced by

$$f(v) \triangleq \sqrt{\langle v, v \rangle}$$

indeed satisfies triangle inequality, thus forming a norm. In addition, given an endomorphism T of some finite-dimensional complex vector space V, by applying Gram-Schmidt process to a Jordan basis, we get an orthogonal basis under which T becomes an upper triangular matrix, called its **Schur's form**.

It should be noted that putting an inner product on a vector space only allow better knowledge. For example, a set of nonzero orthogonal vectors must be linearly independent. This is critical when required to prove an inner product space V has some orthogonal decomposition

$$V = W_1 \oplus \cdots \oplus W_n$$

since the direct sum part would be implied by that they are orthogonal.

Let V, W be two inner product space over $\mathbb{F}$, and let $F : V \to W$ be some linear map. If some linear map $F^\dagger : W \to V$ satisfies

$$\langle Fv, w \rangle = \langle v, F^\dagger w \rangle \quad \text{for all } v \in V, w \in W,$$

we say $F^\dagger$ is an **adjoint** of F. If both V and W are finite-dimensional, then $F^\dagger$ exists uniquely: Given orthogonal bases $\{e_1, \dots, e_n\} \subseteq V, \{f_1, \dots, f_m\} \subseteq W$, the adjoint $F^\dagger$ is uniquely defined by:

$$\overline{B_{j,i}} = \frac{\langle e_i, e_i \rangle}{\langle f_j, f_j \rangle} A_{i,j}, \qquad \text{where } B = [F^\dagger]^{e_1,\dots,e_n}_{f_1,\dots,f_m} \text{ and } A = [F]^{f_1,\dots,f_m}_{e_1,\dots,e_n}$$

Therefore, with respect to orthonormal basis, the matrix representation of $F^\dagger$ is simply the complex conjugation of that of F. It is then obvious that the double adjoint of a linear map between two finite-dimensional inner product spaces over $\mathbb{F}$ is itself.

Given finite dimensional $\mathbb{C}$-vector space V and $T \in \text{End}(V)$, if T is **normal**, i.e. T commutes with its adjoint, then by direct computation, we see that its Schur's form must be diagonal. This result is called **Spectral Theorem for complex finite-dimensional vector space**. If V is over $\mathbb{R}$ instead of $\mathbb{C}$, then T being normal is not enough for T to be orthogonally diagonalized, since we do not have Schur's decomposition in the first place.[3]

Theorem 1.3.1. (Spectral Theorem for real finite-dimensional vector space) Let V be a finite-dimensional real vector space, and let $T \in \text{End}(V)$. If T is **self-adjoint**, then there exists an orthogonal eigenbasis for V with respect to T.

[3]Actually, we do have Schur's decomposition in the sense that we may first decompose a real matrix into a complex Jordan matrix, and then applying the Gram-Schmidt process. Yet, because the matrix representation of adjoint of T when the underlying field is $\mathbb{R}$ is just the transpose of that of T instead of the conjugate transpose, direct computation yield nothing.

Proof. The is proved by induction on the dimension of V. The two base cases correspond to the 0-dimensional and 1-dimensional cases, both of which are trivial. We now prove the inductive case.

Let $v^\perp$ denote the space of vectors orthogonal to v. Because T is self-adjoint, if v is an eigenvector of T, then $v^\perp$ is stable under T. This reduce the problem into proving the existence of an eigenvector.

Let f be some real polynomial that sends T to zero. By Fundamental Theorem of Algebra, we may write

$$0 = f(T) = (T^2 + b_1T + c_1I)\cdots(T^2 + b_nT + c_nI)(T - \lambda_1 I)\cdots(T - \lambda_m I)$$

for some real b_i, c_i, λ_i such that $b_i^2 < 4c_i$ for all i. Because T is self-adjoint, for each i, by Cauchy-Schwarz inequality, we may compute for all $v \neq 0$ that

$$\begin{aligned}
\langle (T^2 + b_iT + c_iI)v, v\rangle &= \langle T^2 v, v\rangle + b_i\langle Tv, v\rangle + c_i\langle v, v\rangle \\
&= \langle Tv, Tv\rangle + b_i\langle Tv, v\rangle + c_i\|v\|^2 \\
&\geq \|Tv\|^2 - b_i\|Tv\| \cdot \|v\| + c_i\|v\|^2 \\
&= \Big(\|Tv\| - \frac{b_i\|v\|}{2}\Big)^2 + \Big(c_i - \frac{b_i^2}{4}\Big)\|v\|^2 > 0
\end{aligned}$$

This implies $T^2 + b_iT + c_iI$ are invertible, which implies for some j, the map $T - \lambda_j I$ is not invertible, i.e., for some j, the real number λ_j is an eigenvalue. ■

Chapter 2

What you probably need to know

2.1 Analytic functions on matrices

Abstract

This section develop the theory of ways to well define values of analytic functions on matrices. TL;DR: If you only care about how to compute e^A, read theorem 2.1.1.

Let $N \in \mathrm{M}_n(\mathbb{C})$ be nilpotent and $\lambda \in \mathbb{C}$. Because λI and N commute, we may apply binomial theorem to see:

$$(\lambda I + N)^m = \sum_{k=0}^{m} \binom{m}{k} \lambda^{m-k} N^k$$

In fact, given $p_d(z) \triangleq a_d(z-z_0)^d + \cdots + a_0 \in \mathbb{C}[z]$, we have

$$\begin{aligned} p_d(\lambda I + N) &= \sum_{m=0}^{d} a_m((\lambda - z_0)I + N)^m \\ &= \sum_{m=0}^{d} a_m \sum_{k=0}^{m} \binom{m}{k} (\lambda - z_0)^{m-k} N^k \\ &= \sum_{k=0}^{d} \left(\sum_{m=k}^{d} a_m \binom{m}{k} (\lambda - z_0)^{m-k} \right) N^k = \sum_{k=0}^{d} \frac{p_d^{(k)}(\lambda)}{k!} N^k \end{aligned}$$

Consider the power series $\sum a_k(z-z_0)^k$ whose convergence disk $B_R(z_0) \subseteq \mathbb{C}$ contains λ.

By basic complex analysis, we know the function $g : B_R(z_0) \to \mathbb{C}$ defined by:

$$g(z) \triangleq \sum_{k=0}^{\infty} a_k (z - z_0)^k$$

is holomorphic and allow term-by-term differentiation. Therefore, we may write:

$$\sum_{k=0}^{\infty} a_k((\lambda - z_0)I + N) = \lim_{d\to\infty} p_d(\lambda I + N) = \lim_{d\to\infty} \sum_{k=0}^{d} \frac{p_d^{(k)}(\lambda)}{k!} N^k = \sum_{k=0}^{\infty} \frac{g^{(k)}(\lambda)}{k!} N^k \tag{2.1}$$

Now, let $U \subseteq \mathbb{C}$ be open and $f : U \to \mathbb{C}$ holomorphic with $\lambda \in U$. Because of equation 2.1, we see that, independent of choice of $z_0 \in U$ that has λ lying in the convergence disk of f at z_0, the series:

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} ((\lambda - z_0)I + N)^k = \sum_{k=0}^{\infty} \frac{f^{(k)}(\lambda)}{k!} N^k$$

always converges to the same matrix. Because of such, *we may define* $f(\lambda I + N) \in M_n(\mathbb{C})$ *with the assurance that every possible computation agree.* Moreover, not only does this definition make philosophical sense, letting $h : U \to \mathbb{C}$ be another holomorphic function, because

$$\frac{(fh)^{(k)}}{k!} = \sum_{r=0}^{k} \binom{k}{r} f^{(r)} h^{(k-r)}, \qquad \text{for all } k \in \mathbb{N} \cup \{0\}$$

we also see the good behavior:

$$f(\lambda I + N)h(\lambda I + N) = (fh)(\lambda I + N)$$

Setting f, h to both be some branches of the multi-valued function $z \mapsto z^{(\frac{1}{r})}$, immediately we see that for all $r \in \mathbb{N}$, there always exists some $A \in M_n(\mathbb{C})$ such that $A^r = \lambda I + N$. (Note that by chain rule, $\sqrt{z}$ always have derivative $\frac{1}{\sqrt{z}}$ where the two $\sqrt{z}$ agree.)

Note that one should be careful in attempt to define value for holomorphic function f on matrix $A \in M_n(\mathbb{C})$ not of the form $\lambda I + N$, since even if the definition can be made precises by considering Jordan form, there may not be a single power series $\sum a_n (z - z_0)^n$ that makes

$$f(A) = \sum a_n (A - z_0 I)^n$$

However, we shall make the remark that **entire functions** are free of this ill situation. Therefore, for every $A \in M_n(\mathbb{C})$, considering its Jordan's form $A = PJP^{-1}$, we see block by block that

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (A - z_0 I)^k = P \cdot \big(f(J(\lambda_1)) \oplus \cdots \oplus f(J(\lambda_n)) \big) \cdot P^{-1}$$

where

$$f(J(\lambda)) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\lambda)}{k!} N^k, \qquad \text{with } J(\lambda) = \lambda I + N$$

The fact that we may generalize entire function $\mathbb{C} \to \mathbb{C}$ to $M_n(\mathbb{C}) \to M_n(\mathbb{C})$ is very important with the following application. Let $A, B \in M_n(\mathbb{F})$ with $\mathbb{F} \in \{\mathbb{C}, \mathbb{R}\}$. The **Frobenius norm** $\|\cdot\|_F$ is defined by:

$$\|A\|_F \triangleq \sum_{i,j} |a_{i,j}|^2 = \operatorname{tr}(A^* A)$$

By Cauchy-Schwarz inequality, we see that the Frobenius norm:

$$\begin{aligned}
\|AB\|^2 &= \sum_{i,j} \Big(\sum_k |A_{i,k} B_{k,j}| \Big)^2 \\
&\leq \sum_{i,j} \Big(\sum_k |A_{i,k}|^2 \Big) \Big(\sum_k |B_{k,j}|^2 \Big) \\
&= \sum_{i,j,k,l} |A_{i,k}|^2 \, |B_{j,l}|^2 = \sum_{i,k} |A_{i,k}|^2 \sum_{j,l} |B_{j,l}|^2 = \|A\|^2 \|B\|^2
\end{aligned}$$

is submultiplicative. Therefore, the **matrix exponential**, which we know take the form

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

make e^{At} converges uniformly in $t \in \mathbb{R}$ on any bounded interval. Because of such, we may perform term-by-term differentiation and get

$$\frac{de^{At}}{dt} = Ae^{At}$$

This implies that for any first order linear system of ODE:

$$\begin{cases} y'(t) = Ay(t) \ \textbf{(DE)} \\ y(0) = y_0 \ \textbf{(IV)} \end{cases}$$

by ODE uniqueness theorem, the solution must have the closed form:

$$y(t) \triangleq e^{At} y_0$$

Moreover, because every general solution to $y' = Ay$ has an initial value at 0, we now see that the general solution space is just the column space of e^{At}. We close this section by showing how to compute the matrix exponential using Jordan form.

Theorem 2.1.1. (Method of computation of matrix exponential) Given square matrix $A \in M_{n\times n}(\mathbb{F})$ with Jordan form:

$$A = QJQ^{-1} = Q(S+N)Q^{-1} \text{ where diagonal } S \text{ and nilpotent } N \text{ commutes.}$$

Because the series $\sum \frac{X^n}{n!}$ always absolutely converge, and because S and N commute, we may copy the argument of Merten's Theorem for Cauchy product to see that

$$e^A = Qe^{S+N}Q^{-1} = Q(e^S e^N)Q^{-1}$$

where both e^S and e^N are easy to compute.

Proof. Skipped. ∎

2.2 Bilinear forms

Let V be a finite dimensional $\mathbb{F}$-vector space, and $f : V^2 \to \mathbb{F}$ be a bilinear form. Clearly, fixing a basis $\{v_1, \dots, v_n\} \subseteq V$, we may express f as a matrix

$$\begin{pmatrix} f(v_1, v_1) & \cdots & f(v_1, v_n) \\ \vdots & \ddots & \vdots \\ f(v_n, v_1) & \cdots & f(v_n, v_n) \end{pmatrix}$$

Changing to a new basis, says $A \triangleq [f]_\alpha$ and $B \triangleq [f]_\beta$, we have

$$A = P^t BP, \quad \text{where } \alpha_i = \sum_{j=1}^n P_{i,j}\beta_j$$

This allows us to define **rank** and **degeneracy** for bilinear forms. Moreover, it allow us to define whether f is **symmetric** or **skew-symmetric**. We immediately see f is symmetric if and only if $f(v, w) = f(w, v)$ for all $w, v \in V$. We also see f is skew-symmetric if and only if $f(v, w) = -f(w, v)$ for all $w, v \in V$.

For each symmetric bilinear form $f : V^2 \to \mathbb{F}$, we associate it with a **quadratic form** $q : V \to \mathbb{R}$ by

$$q(v) \triangleq f(v, v)$$

Such association is injective since

$$f(v, w) = \frac{q(v + w) - q(v - w)}{4}$$

Theorem 2.2.1. (Decomposition theorem for skew-symmetric bilinear form of finite dimensional real vector space) Let $A \in M_n(\mathbb{R})$ be skew-symmetric. Because $A^t A \in M_n(\mathbb{R})$ is symmetric, by spectral theorem for finite dimensional real vector space, we have an orthogonal decomposition:

$$\mathbb{R}^n \cong E(A^t A, \lambda_1) \oplus \cdots \oplus E(A^t A, \lambda_n) \oplus E(A^t A, 0), \quad \text{where } \lambda_i \neq 0$$

in which $\lambda_i > 0$ and $\dim (E(A^t A), \lambda_i) = 2$. These eigenspaces are all invariant under A. Moreover, there exists orthonormal basis of $\mathbb{R}^n$ that makes the matrix representations of restrictions of A become:

$$\begin{bmatrix} 0 & \sqrt{\lambda_i} \\ -\sqrt{\lambda_i} & 0 \end{bmatrix} \quad \text{and} \quad 0$$

Proof. Fix i. Let $v_1 \in E(A^tA, \lambda_i)$ be unital. The statement $\lambda_i > 0$ follows from

$$|Av_1|^2 = (Av_1)^t Av_1 = v_1^t A^t A v_1 = \lambda_i |v_1|^2$$

We now prove that $E(A^tA, \lambda_i)$ is invariant under A. Let $v_2 = \frac{1}{-\sqrt{\lambda_i}} Av_1$. Indeed, we see:

$$A^tAv_2 = \frac{1}{-\sqrt{\lambda_i}} A^tAAv_1 = \frac{1}{-\sqrt{\lambda_i}} AA^tAv_1 = \frac{1}{-\sqrt{\lambda_i}} \cdot \lambda_i Av_1 = -\sqrt{\lambda_i} v_1 = \lambda_i v_2$$

Clearly v_2 is also unital. Because

$$\langle v_1, Av_1 \rangle = \langle A^t v_1, v_1 \rangle = \langle v_1, -Av_1 \rangle \text{ implies } \langle v_1, Av_1 \rangle = 0$$

We see that indeed v_1 and v_2 are orthogonal. Similar technique then go on to show that $Av_2 = \sqrt{\lambda_1} v_1$ so the matrix representation of restriction of A with respect to $\{v_1, v_2\}$ has the desired form. Lastly, since

$$A^tAw = 0 \implies |Aw|^2 = w^tA^tAw = 0$$

we see that A indeed have trivial action on $E(A^tA, 0)$. ■

2.3 SVD

Let V, W be two finite-dimensional inner product space over $\mathbb{F}$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Given $F \in \mathrm{Hom}(V, W)$, clearly $F^\dagger F \in \mathrm{End}(V)$ is self-adjoint. Therefore, by Spectral Theorem, there exists some orthonormal eigenbasis α of V with respect to $F^\dagger F$. Because for every pair $e_1, e_2 \in \alpha$, whether $F(v_1), F(v_2)$ are both nonzero or not, we have:

$$\langle F(e_1), F(e_2) \rangle = \langle e_1, F^\dagger F(e_2) \rangle = 0$$

By deleting the zeros and negation of some elements, we may extend $\{F(e) \in W : e \in \alpha\}$ to an orthonormal basis β for W such that every non-zero entry of the $\dim(V)$-by-$\dim(W)$ $\mathbb{C}$-matrix $[F]_\alpha^\beta$ is a positive real number and lies in the diagonal. This process is called the **singular value decomposition of F**.

Chapter 3

What won't hurt to know

3.1 Powers of Jordan blocks

In this section, we denote Jordan block by $J(\lambda, n)$. For example, by $J(\sqrt{5}, 2)$ we mean:

$$\begin{pmatrix} \sqrt{5} & 1 \\ 0 & \sqrt{5} \end{pmatrix}$$

This section only prove one result:

Theorem 3.1.1. (Powers of Jordan blocks) Let $\mathbb{F}$ has zero characteristic, $\lambda \in \mathbb{F}$, and $n, t \in \mathbb{N}$. Then:

$$(J(\lambda, n))^t \text{ has the Jordan form: } \begin{cases} J(\lambda^t, n) & \text{if } \lambda \neq 0 \\ \begin{pmatrix} J(0, n-t+1) & 0 \\ 0 & 0 \end{pmatrix} & \text{if } \lambda = 0 \text{ and } t < n \\ 0 & \text{if } \lambda = 0 \text{ and } t \geq n \end{cases}$$

Proof. (iii) is clear from computation. (ii) is also clear from computation. Indeed, if $t < n$, then:

$$J(0, n)^t = P \begin{pmatrix} J(0, n-t+1) & 0 \\ 0 & 0 \end{pmatrix} P^{-1}$$

where

$$P = \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_{t+1} & \mathbf{e}_{t+2} & \cdots & \mathbf{e}_n & \mathbf{e}_2 & \mathbf{e}_3 & \cdots \mathbf{e}_t \end{bmatrix}$$

We now prove (i). For simplicity, from now on we write J in place of $J(\lambda, n) = I + \lambda N$. Clearly, J^t is always an upper triangular matrix whose diagonal entries are all λ^t, and its eigenvalues in $\overline{\mathbb{F}}$ are all λ^t. We are required to prove that J^t has only one Jordan block, i.e., $(J^t - \lambda^t I)^{n-1} \neq 0$. Using binomial theorem, we may compute:

$$J^t = (N + \lambda I)^t = \sum_{k=0}^{t} \binom{t}{k} \lambda^{t-k} N^k$$

This implies:

$$J^t - \lambda^t I = a_{n-1} N^{n-1} + \cdots + a_1 N^1, \quad \text{where } a_i \triangleq \binom{t}{i} \lambda^{t-i}$$

which implies:

$$(J^t - \lambda^t I)^{n-1} = a_1^{n-1} N^{n-1} \neq 0$$

■

Clearly, theorem 3.1.1 give us the **spectral theorem for polynomials**, whatever that means to you.

3.2 Gram matrix

Abstract

Let $\mathbb{F}$ be a subfield of $\mathbb{C}$. This section use Gram matrix to establish a correspondence between the set of positive-definite matrix in $\mathrm{M}_n(\mathbb{F})$ and the set of inner products on $\mathbb{F}^n$.

Let $M \in \mathrm{M}_n(\mathbb{F})$. We say M is **Hermitian** if $M = M^*$. If M is Hermitian, clearly $\mathbf{x}^* M \mathbf{x} \in \mathbb{C}$ must be real for all $\mathbf{x} \in \mathbb{F}^n$. We say M is **positive-definite** if M is Hermitian and $\mathbf{x}^* M \mathbf{x} > 0$ for all nonzero $\mathbf{x} \in \mathbb{F}^n$.

Let W be an inner product $\mathbb{F}$-vector space, and let $\{w_1, \dots, w_n\}$ be a basis. The matrix $G \in M_n(\mathbb{F})$ defined by $G_{i,j} \triangleq \langle w_i, w_j \rangle$ is called the **Gram matrix** of W. G is clearly Hermitian. To see it is moreover positive-definite, simply compute

$$\mathbf{x}^* G \mathbf{x} = \left\langle \sum_i \overline{x_i} w_i, \sum_i \overline{x_i} w_i \right\rangle > 0, \quad \text{for all } \mathbf{x} \in \mathbb{F}^n$$

Note that clearly we have

$$\left\langle \sum \overline{c_i} w_i, \sum \overline{d_i} w_i \right\rangle = \mathbf{c}^* G \mathbf{d}$$

Note that if $M \in \mathrm{M}_n(\mathbb{F})$ is positive definite, then M define an inner product on $\mathbb{F}^n$ by

$$\langle \mathbf{x}, \mathbf{y} \rangle_M = \mathbf{y}^* M \mathbf{x}$$

3.3 Simultaneous Triangulation

Theorem 3.3.1. (triangulable if and only if minimal polynomial splits if and only if Jordan form exists if and only if one step extension property) Let V be a finite dimensional $\mathbb{F}$-vector space, and let $T \in \text{End}(V)$. Then the followings are equivalent:

(i) T is triangulable.

(ii) minimal polynomial of T splits.

(iii) For every proper subspace W, there exists some $v \in V - W$ such that Tv is in the span of W and v.

Proof. (i) $\iff$ (ii) is clear. (iii) $\implies$ (i) is also clear. We now show (i) $\implies$ (iii). Let $\{e_1, \dots, e_n\}$ be a basis that upper triangularize T. Because W is proper, we may let $i \in \{1, \dots, n\}$ be the smallest index such that $e_i \notin W$. Such e_i then clearly suffice. ■

Theorem 3.3.2. (One step extension property of commuting family of triangulable operators) Let V be a finite dimensional $\mathbb{F}$-vector space, and let elements of $\mathcal{F} \subseteq \text{End}(V)$ all be triangulable. If $\mathcal{F}$ commute and W is a proper $\mathcal{F}$-invariant subspace, then there is some $v \in V - W$ such that Tv is in the span of W and v for all $T \in \mathcal{F}$.

Proof. Clearly, we may suppose $\mathcal{F}$ is finite $\{T_1, \dots, T_n\}$. By theorem 3.3.1, there exists some $v_1 \in V - W$ and $c_1 \in \mathbb{F}$ such that $(T_1 - c_1 I)v_1 \in W$. Therefore, the vector space $V_1 \triangleq \{v \in V : (T_1 - c_1 I)v \in W\}$ is strictly larger than W.

Because W is $\mathcal{F}$-invariant, we see V_1 is also $\mathcal{F}$-invariant. Clearly $T_2|_{V_1}$ is also triangulable. Therefore by theorem 3.3.1, there exists some $v_2 \in V_1 - W$ and $c_2 \in \mathbb{F}$ such that $(T_2 - c_2 I)v_2 \in W$. Let $V_2 \triangleq \{v \in V_1 : (T_1 - c_2 I)v \in W\}$. Using the same method, we may define V_n. We now see that any vector in $V_n - W$ suffices. ■

Corollary 3.3.3. (Commuting family of triangulable operators is simultaneously triangulable) Let V be a finite dimensional $\mathbb{F}$-vector space, and let elements of $\mathcal{F} \subseteq \text{End}(V)$ all be triangulable. If $\mathcal{F}$ commute it is simultaneously triangulable.

3.4 Cauchy interlacing theorem

Theorem 3.4.1. (Cauchy interlacing theorem) Let $A \in M_n(\mathbb{R})$ be symmetric with eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$ and $B \in M_m(\mathbb{R})$ a principal submatrix of A with eigenvalues $\beta_1 \leq \cdots \leq \beta_m$. For all β_k, we have:

$$\lambda_k \leq \beta_k \leq \lambda_{n+k-m}$$

In particular, when $m = n-1$, we have

$$\lambda_1 \leq \beta_1 \leq \lambda_2 \leq \cdots \leq \lambda_{n-1} \leq \beta_{n-1} \leq \lambda_n$$

Proof. WLOG, write

$$A = \begin{pmatrix} B & X \\ X^t & Z \end{pmatrix}$$

Let A has eigenbasis $\{x_1, \dots, x_n\}$ that corresponds to eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$, and let B has eigenbasis $\{y_1, \dots, y_m\}$ that corresponds to eigenvalues $\beta_1 \leq \cdots \leq \beta_m$. Consider

$$V \triangleq \text{span}(x_k, \dots, x_n) \quad \text{and} \quad W \triangleq \text{span}(y_1, \dots, y_k)$$

By counting dimension, we see that there exists $w \in W$ such that

$$\widetilde{w} \triangleq \begin{pmatrix} w \\ 0 \end{pmatrix} \in V \quad \text{and} \quad |w| = 1$$

Compute that

$$\widetilde{w}^t A \widetilde{w} = \begin{pmatrix} w \\ 0 \end{pmatrix}^t \begin{pmatrix} B & X \\ X^t & Z \end{pmatrix} \begin{pmatrix} w \\ 0 \end{pmatrix} = w^t B w$$

We now see

$$\lambda_k \leq \widetilde{w}^t A w = w^t B w \leq \beta_k$$

Same method can be applied to show $\beta_k \leq \lambda_{n+k-m}$ by setting

$$V \triangleq \text{span}(x_1, \dots, x_{n+k-m}) \quad \text{and} \quad W \triangleq \text{span}(y_k, \dots, y_m)$$

■

Theorem 3.4.2. (Cauchy interlacing theorem) Let $A \in M_n(\mathbb{C})$ be Hermitian, and $B \in M_m(\mathbb{C})$ a principal submatrix of A. Because they are Hermitian, we know their eigenvalues are real. Let A has eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$ and B has eigenvalues $\beta_1 \leq \cdots \leq \beta_m$. For all β_k, we have:

$$\lambda_k \leq \beta_k \leq \lambda_{n+k-m}$$

In particular, when $m = n-1$, we have

$$\lambda_1 \leq \beta_1 \leq \lambda_2 \leq \cdots \leq \lambda_{n-1} \leq \beta_{n-1} \leq \lambda_n$$

Proof. This is proved exactly the same way. ■

Corollary 3.4.3. (Sylvester's Criterion) Let $A \in M_n(\mathbb{C})$ be Hermitian or $A \in M_n(\mathbb{R})$ be symmetric. The followings are equivalent:

(i) A is positive definite.

(ii) Every leading principal minor (including A itself) is positive.

3.5 When is T a constant multiple of the identity?

Theorem 3.5.1. (When T preserves every subspace) Let V be a finite dimensional $\mathbb{F}$-vector space, and let $T \in \mathrm{End}(V)$. If every subspace of V is T-invariant, then T is a constant multiple of identity.

Proof. Clearly every vector is an eigenvector. If T has distinct eigenvalues, says $Tv = 2v$ and $Tw = w$, then clearly we have a contradiction, namely that $v + w$ is not an eigenvector. ■

Theorem 3.5.2. (When T commutes with every projection) Let V be a finite dimensional $\mathbb{F}$-vector space, and let $T \in \mathrm{End}(V)$. If T commutes with every projection, then T is a constant multiple of identity.

Proof. If T commute with projection $E \in \mathrm{End}(V)$, then clearly both image and kernel of E is T-invariant. This together with theorem 3.5.1 suffices. ■

Theorem 3.5.3. (When T commute with every diagonalizable operator) Let V be a finite dimensional $\mathbb{F}$-vector space, and let $T \in \mathrm{End}(V)$. If T commutes with every diagonalizable operator, then T is a constant multiple of identity.

Proof. This follows from theorem 3.5.2 and noting that projections are diagonalizable. ■

3.6 Dimension Formula for Centralizer of Matrices that admits Jordan Form

Let $\mathbb{F}$ be a field and $A \in M_n(\mathbb{F})$. Clearly, its **centralizer**

$$C(A) \triangleq \{B \in M_n(\mathbb{F}) : AB = BA\}$$

forms a $\mathbb{F}$-vector subspace of $M_n(\mathbb{F})$. If A admits an Jordan form in $\mathbb{F}$, we may give a formula–which is the only result we prove in this section–for the dimension of $C(A)$ depending on it:

Theorem 3.6.1. (Dimension of centralizer of matrix that has Jordan form) If $A \in M_n(\mathbb{F})$ has the Jordan form:

$$A = J(\lambda_1, d_{1,1} \oplus \cdots \oplus d_{1,k_1}) \oplus \cdots \oplus J(\lambda_t, d_{t,1} \oplus \cdots \oplus d_{t,k_t}), \quad \lambda_i \in \mathbb{F}$$

where $d_{i,1} \geq \cdots \geq d_{i,k_i}$ for all i. Then:

$$\dim(C(A)) = \sum_{l=1}^{t} \sum_{i=1}^{k_l} \sum_{j=1}^{k_l} \min\{d_{l,i}, d_{l,j}\}$$

Or equivalently:

$$\dim(C(A)) = \sum_{i=1}^{t} \sum_{j=1}^{k_i} (2j-1)(d_{i,j}-1)$$

Proof. Denote $V \triangleq \mathbb{F}^n$ and $R \triangleq \mathbb{F}[x]$. Give V the natural R-module structure by letting $T \cdot v \triangleq Av$. Clearly, $C(A)$ is isomorphic to $\operatorname{End}_R(V)$ as a $\mathbb{F}$-vector space. Because

$$\operatorname{End}_R(V) \cong \bigoplus_{(\lambda,d),(\mu,e)} \operatorname{Hom}_R(J(\lambda,d), J(\mu,e))$$

as R-module, we only have to show:

(I) $\operatorname{Hom}_R(J(\lambda,d), J(\mu,e)) = 0, \quad$ if $\lambda \neq \mu$.

(II) $\dim_{\mathbb{F}}(\operatorname{Hom}_R(J(\lambda,d), J(\lambda,e))) = \min\{d,e\}$.

(III) $\sum_{i=1}^{k_l} \sum_{j=1}^{k_l} \min\{d_{l,i}, d_{l,j}\} = \sum_{j=1}^{k_l} (2j-1)(d_{l,j}-1), \quad$ for all $l \in \{1,\dots,t\}$.

The purely combinatorial fact (III) is easy to check. We now prove (I). Let $v \in J(\lambda,d)$ and $f \in \operatorname{Hom}_R(J(\lambda,d), J(\mu,e))$. We are required to prove $f(v) = 0$. Let $\{w_1,\dots,w_e\} \subseteq V$ be the Jordan basis for $J(\mu, e)$, i.e.,

$$\left[A|_{J(\mu,e)}\right]_{\{w_1,\dots,w_e\}} = \begin{pmatrix} \mu & 1 & & \\ & \mu & 1 & \\ & & \ddots & \ddots \\ & & & \mu \end{pmatrix}$$

We may uniquely write $f(v) = \sum c_i w_i$. Because f is a R-module homomorphism, we know:

$$\sum c_i (A - \lambda I)^d w_i = f((A - \lambda I)^d v) = f(0) = 0 \tag{3.1}$$

Let $\mathcal{M}$ denote the $e \times e$-matrix:

$$\left[(A - \lambda I)^d|_{J(\mu,e)}\right]_{\{w_1,\dots,w_e\}}$$

Because $\mathcal{M}$ is upper triangular whose diagonals are all nonzero[1], equation 3.1 implies $c_e = 0$. Repeating the same process, one see $c_i = 0$ for all i, proving (I). It remains to prove (II). Denote $t \triangleq \min\{d, e\}$, and let $\{v_1, \dots, v_d\}$ and $\{w_1, \dots, w_e\}$ respectively be the Jordan bases of $J(\lambda, d)$ and $J(\lambda, e)$. Let $\mathcal{N} \subseteq M_{e\times d}(\mathbb{F})$ be the $\mathbb{F}$-vector subspace of matrices of the form:

$$\begin{pmatrix} \mathcal{O}_1 & \mathcal{K} \\ \mathcal{O}_2 & \mathcal{O}_3 \end{pmatrix}$$

where t-by-$(d-t)$ matrix $\mathcal{O}_1$, $(e-t)$-by-$(d-t)$ matrix $\mathcal{O}_2$, and $(e-t)$-by-t matrix $\mathcal{O}$ are all zero, while $t \times t$-matrix $\mathcal{K}$ is upper triangular with $\mathcal{K}_{i,j} = \mathcal{K}_{i+k,j+k}$ for all i, j, k.

To show $\dim_{\mathbb{F}} (\mathrm{Hom}_R(J(\lambda, d)), J(\lambda, e)) = t$, we only have to show that the $\mathbb{F}$-vector space homomorphism $\phi : \mathrm{Hom}_R (J(\lambda, d), J(\lambda, e)) \to M_{d\times e}(\mathbb{F})$ defined by

$$\phi(f) \triangleq [f]^{\{w_1,\dots,w_e\}}_{\{v_1,\dots,v_d\}}$$

forms a $\mathbb{F}$-vector space isomorphism between $\mathrm{Hom}_R (J(\lambda, d), J(\lambda, e))$ and $\mathcal{N}$. This requires proving three statements:

(S_1) ϕ is injective.

(S_2) $\mathrm{Im}(\phi) \subseteq \mathcal{N}$.

(S_3) $\mathrm{Im}(\phi) \supseteq \mathcal{N}$.

[1]They are all $(\mu - \lambda)^d$.

The injectivity of ϕ is clear from the definition. We now show $\mathrm{Im}(\phi) \subseteq \mathcal{N}$. Let $f \in \mathrm{Hom}_R\left(J(\lambda, d), J(\lambda, e)\right)$. Write $f(v_d) \triangleq \sum b_i w_i$. Because f is a R-module homomorphism, we know for all $i \leq d-1$ that:

$$f(v_{d-i}) = f\left((A-\lambda I)^i v_d\right) = (A-\lambda I)^i f(v_d) = \sum_{j=1}^{e} b_j (A-\lambda I)^i w_j = \sum_{j=i+1}^{e} b_j w_{j-i} \tag{3.2}$$

Moreover, since $(A-\lambda I)^d v_d = 0$, we also know

$$\sum_{i=1}^{e-d} b_{i+d} w_i = (A-\lambda I)^d f(v_d) = f\left((A-\lambda I)^d v_d\right) = 0$$

which implies $b_{t+i} = 0$ for all i.[2] This together with the fact that equation 3.2 holds for all $i \leq d-1$ shows that $\mathrm{Im}(\phi) \subseteq \mathcal{N}$. Let $f \in \mathrm{Hom}_{\mathbb{F}}\left(J(\lambda, d), J(\lambda, e)\right)$ satisfies:

$$[f]_{\{v_1,\dots,v_d\}}^{\{w_1,\dots,w_e\}} \in \mathcal{N}$$

To show $\mathrm{Im}(\phi) \supseteq \mathcal{N}$, we only have to show that f lies in $\mathrm{Hom}_R\left(J(\lambda, d), J(\lambda, e)\right)$, i.e., f is moreover a R-module homomorphism. To prove that f forms a R-module homomorphism, clearly we only have to prove $f \circ A|_{J(\lambda,d)} = A|_{J(\lambda,e)} \circ f$, which can be easily checked by direct matrix computation. ■

[2] If $d \geq e$, then the statement $b_{t+i} = 0$ for all i holds true trivially.

3.7 Cyclic Vector Spaces

Equivalent Definition 3.7.1. (cyclic vector space) Let V be a finite dimensional $\mathbb{F}$-vector space, and $T \in \operatorname{End}(V)$. Then the followings are equivalent:

(i) T is **cyclic**, i.e., there exists some v such that $V = \operatorname{span}\{T^n v : n \in \mathbb{N} \cup \{0\}\}$.

(ii) Minimal polynomial m_T of T equals to characteristic polynomial c_T.

(iii) Centralizer $C(T)$ of T equals to $\mathbb{F}[T]$.

Proof. (i) $\iff$ (ii) follows from invariant factor form of structure theorem for finitely generated modules over PID. Clearly we have (i) $\implies$ (iii). We now prove (iii) $\implies$ (i). Assume T isn't cyclic for a contradiction. Then we may write

$$V \cong \mathbb{F}[x]/(a_1) \oplus \cdots \oplus \mathbb{F}[x]/(a_n) \oplus \mathbb{F}[x]/(m_T)$$

where $a_n \mid m_T$. Consider the $\mathbb{F}[x]$-module endomorphism that maps $(0, \ldots, 1)$ to $(0, \ldots, 1)$. Because such $\mathbb{F}[x]$-module homomorphism is a polynomial in T, we know it must be of the form $qm_T + 1$. However, if we require it to maps $(0, \ldots, 1, 0)$ to $(0, \ldots, 0, 0)$, then we see a contradiction. ■

Chapter 4

Multi Linear Algebra

4.1 Tensor Product

In the most general sense, the **tensor product** of two $\mathbb{F}$-vector space V_1, V_2 is merely a $\mathbb{F}$-vector space X associated with a $\mathbb{F}$-bilinear map $g : V_1 \times V_2 \to X$ that satisfies the **universal property** of tensor product: For every $\mathbb{F}$-bilinear map $f : V_1 \times V_2 \to W$, there exists a unique $\mathbb{F}$-linear map $\widetilde{f} : X \to W$ such that $f = \widetilde{f} \circ g$.

It is routine to check that the universal property of tensor product defines vector space up to an isomorphism compatible with itself. More precisely, if $Y, h : V_1 \times V_2 \to Y$ also satisfies the universal property, then there exists an isomorphism $f : X \to Y$ such that $f \circ g = h$.

Because of such, we often write X simply as $V_1 \otimes V_2$ and denote $g(v_1, v_2)$ by $v_1 \otimes v_2$. One common construction of the tensor product is to define $V_1 \otimes V_2$ as the space of $\mathbb{F}$-bilinear maps from $V_1^\vee \times V_2^\vee$ to $\mathbb{F}$ and to define the $\mathbb{F}$-bilinear map by $(v_1 \otimes v_2)(\varphi_1, \varphi_2) \triangleq \varphi_1(v_1)\varphi_2(v_1)$.

At this point, one may already observe that we may restate the universal property of tensor product as follows. Given a finite collection $\{V_1, \dots, V_n\}$ of $\mathbb{F}$-vector spaces, we say the $\mathbb{F}$-vector space $V_1 \otimes \cdots \otimes V_n$ and the $\mathbb{F}$-multilinear map $\otimes : V_1 \times \cdots \times V_n \to V_1 \otimes \cdots \otimes V_n$ satisfies the universal property if for every $\mathbb{F}$-multilinear map $f : V_1 \times \cdots \times V_n \to W$, there exists a unique $\mathbb{F}$-linear map $\widetilde{f} : V_1 \otimes \cdots \otimes V_n$ such that $\widetilde{f}(v_1 \otimes \cdots \otimes v_n) = f(v_1, \dots, v_n)$ for every $(v_1, \dots, v_n) \in V_1 \times \cdots \times V_n$.

Even thought the construction of the tensor product is deemed unimportant by general consensus, here, we make the remark that the construction we gave do show that if B_j are finite bases for V_j, then $\{v_1 \otimes \cdots \otimes v_n \in V_1 \otimes \cdots \otimes V_n : v_j \in B_j \text{ for all } j\}$ forms a basis for $V_1 \otimes \cdots \otimes V_n$. In fact, this together with the universal property may be the only things about tensor product one will ever need to develop other properties of tensor products.

4.2 Exterior Algebra

Let V be some finite-dimensional $\mathbb{F}$-vector space. In this note, by the **exterior n-th power** $\bigwedge^n V$ of V, we mean the space of the antisymmetric multilinear maps from $(V^*)^n$ to $\mathbb{F}$, together with a multilinear map $\wedge : V^n \to \bigwedge^n V$ defined by

$$(v_1 \wedge \cdots \wedge v_n)(\varphi_1, \ldots, \varphi_n) \triangleq \begin{vmatrix} \varphi_1 v_1 & \cdots & \varphi_1 v_n \\ \vdots & \ddots & \vdots \\ \varphi_n v_1 & \cdots & \varphi_n v_n \end{vmatrix}$$

One may check that

(a) $\wedge$ is itself antisymmetric.

(b) If $\{v_1, \ldots, v_k\}$ forms a basis for V, then $\{v_{i_1} \wedge \cdots \wedge v_{i_n} \in \bigwedge^n V : 1 \leq i_1 < \cdots < i_n \leq k\}$ forms a basis for $\bigwedge^n V$.

(c) $\bigwedge^n V$ also satisfies its **universal property**: For every antisymmetric multilinear map $f : V^n \to W$, there exists a unique linear map $\widetilde{f}$ such that $\widetilde{f} = f \circ \wedge$.

Because of the universal property, after some tedious construction, we see that there exists some unique bilinear map $g : \bigwedge^n V \times \bigwedge^k V$ such that $g(v_1 \wedge \cdots \wedge v_n, v_{n+1} \wedge \cdots \wedge v_{n+k}) = v_1 \wedge \cdots \wedge v_{n+k}$ for all $(v_1, \ldots, v_{n+k}) \in V^{n+k}$. Therefore, when we define the **exterior algebra of** V by

$$\bigwedge V \triangleq \bigoplus_{n=0}^{\infty} \bigwedge^n V$$

where $\bigwedge^0 V = \mathbb{F}$ by convention, we see that $\bigwedge V$ indeed have the structure of an associative $\mathbb{F}$-algebra where the vector multiplication $\wedge : \bigwedge V \times \bigwedge V \to \bigwedge V$ satisfies

$$(v_1 \wedge \cdots \wedge v_n) \wedge (v_{n+1} \wedge \cdots \wedge v_{n+k}) = v_1 \wedge \cdots \wedge v_{n+k} \tag{4.1}$$

for all $(v_1, \ldots, v_{n+k}) \in V^{n+k}$. Again, since both exterior power and exterior algebra are categorical concept, the fore tedious construction doesn't really matter. What matter is the property they always have, for example Equation 4.1, the fact wedge products is antisymmetric, or the fact $\{v_{i_1} \wedge \cdots \wedge v_{i_n} \in \bigwedge^n V : 1 \leq i_1 < \cdots < i_n \leq k\}$ forms a basis for $\bigwedge^n V$. For future references, we close this section by giving two other important properties of the exterior algebra.

(a) For arbitrary $\sigma \in S_n$ and arbitrary $e_1, \ldots, e_n \in V$, we have

$$e_{\sigma(1)} \wedge \cdots \wedge e_{\sigma(n)} = \text{sgn}(\sigma) e_1 \wedge \cdots \wedge e_n$$

(b) For arbitrary $e_1, \ldots, e_n \in V$ and arbitrary $A \in M_n(\mathbb{F})$, we have

$$f_i = \sum_j A_{i,j} e_j \text{ for all } i \implies f_1 \wedge \cdots \wedge f_n = \det(A) e_1 \wedge \cdots \wedge e_n$$

4.3 Pfaffian

Given an $2n \times 2n$ skew-symmetric matrix A over $\mathbb{F}$, we define its **Pfaffian** to be

$$\operatorname{pf}(A) \triangleq \frac{1}{2^n n!} \sum_{\sigma \in S_{2n}} \operatorname{sgn}(\sigma) \prod_{k=1}^{n} A_{\sigma(2k-1),\sigma(2k)}$$

Theorem 4.3.1. (Equivalent Definition of Pfaffian) If we define $\omega \in \bigwedge^{2n} \mathbb{F}$ by

$$\omega \triangleq \sum_{i,j} A_{i,j} e_i \wedge e_j$$

then

$$\omega^n = 2^n n! \operatorname{pf}(A) e_1 \wedge e_2 \wedge \cdots \wedge e_{2n}$$

Proof. Let X be the set of functions that maps $\{1, \dots, n\}$ into $\{1, \dots, 2n\} \times \{1, \dots, 2n\}$ such that when we write $f(k) = (f_1(k), f_2(k))$ for $f \in X$, we have

$$\{f_1(k) : 1 \leq k \leq n\} \cup \{f_2(k) : 1 \leq k \leq n\} = \{1, \dots, 2n\}$$

Because the wedge product is antisymmetric, we have

$$\omega^n = \sum_{f \in X} \Big(\prod_{k=1}^{n} A_{f_1(k), f_2(k)} \Big) e_{f_1(1)} \wedge e_{f_2(1)} \wedge \cdots \wedge e_{f_n(1)} \wedge e_{f_n(2)}$$

It remains to show

$$\sum_{f \in X} \Big(\prod_{k=1}^{n} A_{f_1(k), f_2(k)} \Big) e_{f_1(1)} \wedge e_{f_2(1)} \wedge \cdots \wedge e_{f_n(1)} \wedge e_{f_n(2)}$$
$$= \sum_{\sigma \in S_{2n}} \operatorname{sgn}(\sigma) \Big(\prod_{k=1}^{n} A_{\sigma(2k-1),\sigma(2k)} \Big) e_1 \wedge \cdots \wedge e_{2n}$$

Define $\phi : S_{2n} \to X$ by

$$(\phi(\sigma))(k) \triangleq (\sigma(2k-1), \sigma(2k))$$

Clearly, ϕ is a bijection, and

$$\prod_{k=1}^{n} A_{\sigma(2k-1),\sigma(2k)} = \prod_{k=1}^{n} A_{\phi(\sigma)_1(k), \phi(\sigma)_2(k)}$$

and

$$e_{\phi(\sigma)_1(1)} \wedge e_{\phi(\sigma)_2(1)} \wedge \cdots \wedge e_{\phi(\sigma)_1(n)} \wedge e_{\phi(\sigma)_2(n)} = e_{\sigma(1)} \wedge e_{\sigma(2)} \wedge \cdots \wedge e_{\sigma(2n)} = \operatorname{sgn}(\sigma) e_1 \wedge \cdots \wedge e_{2n}$$

Above equation need to be explained in the section before. ■

Chapter 5

Solutions to NTU Mathematics Master Program Entrance Exams

5.1 Year 114

Question 1

Let

$$A \triangleq \begin{pmatrix} 2 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

Find the Smith normal form of $xI - A \in M_4(\mathbb{R}[x])$ and the Jordan form of A.

Proof. The solution is that $\mathrm{SNF}_{\mathbb{R}[x]}(xI - A) = \operatorname{diag}\left\{1, 1, x-1, (x-1)^2(x-2)\right\}$ and that

$$J \triangleq \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{ and } P \triangleq \begin{pmatrix} -1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & -1 & -1 & 0 \\ -1 & 0 & 1 & 0 \end{pmatrix}$$

makes $A = PJP^{-1}$. ■

Question 2

Let

$$A \triangleq \begin{pmatrix} 3 & -2 & -2 \\ -2 & 0 & 1 \\ -2 & 1 & 0 \end{pmatrix}$$

(i) Find the maximum of $\mathbf{x}^t A\mathbf{x}$ among all $\mathbf{x} \in \mathbb{R}^3$ that makes $|\mathbf{x}| = 1$.

(ii) Find the minimum of $Y^t AY$ among all $Y \in M_{3\times 2}(\mathbb{R})$ that makes $Y^t Y = I$.

Proof. Because A is symmetric, by spectral theorem for finite dimensional real vector space, we know there exists an orthonormal eigenbasis

$$\left\{ \frac{1}{\sqrt{6}} \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$$

Denote them in order by $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and let $\mathbf{x} = c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + c_3\mathbf{e}_3$. Because they form an orthonormal eigenbasis, we know

$$\mathbf{x}^t A\mathbf{x} = 5c_1^2 - c_2^2 - c_3^2$$

This implies that the solution for (i) is -5 with $\mathbf{x} = \mathbf{e}_1$ and the solution for (ii) is -2 with $Y = \begin{pmatrix} \mathbf{e}_2 & \mathbf{e}_2 \end{pmatrix}$. ■

Question 3

Let $A \in M_n(\mathbb{C})$ be invertible. Show that $A^{-1} = p(A)$ for some $p(x) \in \mathbb{C}[x]$.

Proof. We are required to prove the existence of $p(x) \in \mathbb{C}[x]$ such that $p(x) \cdot x - 1 \in \mathbb{C}[x]$ maps A to 0. Because A is invertible, by looking at the Jordan form of A, we know the minimal polynomial m of A has nonzero constant term. Let the constant term of m be -1. We then see $p \triangleq \frac{m+1}{x}$ suffices. ■

Question 4

Let V and W be two finite dimensional inner product space over $\mathbb{R}$.

(i) Prove that for each $w \in W$, there exists a unique vector in V denoted by T^*w such that $\langle Tv, w\rangle = \langle v, T^*w\rangle$ for all $v \in V$. Show that T^* is linear.

(ii) Prove that $(\operatorname{Im} T)^\perp = \ker T^*$.

(iii) Prove that $\dim\ker T - \dim\ker T^* = \dim V - \dim W$.

Proof. Let $\{e_1, \dots, e_n\}, \{u_1, \dots, u_m\}$ be orthonormal basis of V, W with $\{u_1, \dots, u_k\}$ be a basis for $\operatorname{Im} T$. Fix $w \in W$. Write

$$T^* w = c_1 e_1 + \cdots + c_n e_n$$

Because e_i are orthonormal, by premise, we can only have $c_i = \langle Te_i, w\rangle$. We have shown the uniqueness of T^*w. To see such T^*w suffices, simply compute

$$\left\langle T \sum d_i e_i, w \right\rangle = \sum d_i \langle Te_i, w\rangle = \sum d_i c_i = \left\langle \sum d_i e_i, \sum c_i e_i \right\rangle = \left\langle \sum d_i e_i, T^* w \right\rangle$$

To see T^* is linear, just observe

$$\langle Te_i, w_1 + w_2\rangle = \langle Te_i, w_1\rangle + \langle Te_i, w_2\rangle$$

Clearly we have $(\operatorname{Im} T)^\perp \supseteq \ker T^*$. Fix $i > k$. To prove the other direction, we are required to prove $u_i \in \ker T^*$. Because $\{u_1, \dots, u_k\}$ spans $\operatorname{Im} T$ and $\{u_1, \dots, u_m\}$ forms an orthonormal basis, we may compute

$$T^* u_i = \langle Te_1, u_i\rangle e_1 + \cdots + \langle Te_n, u_i\rangle e_n = 0$$

as desired.

Fix $1 \le i \le m$. Compute

$$T^* u_i = \langle Te_1, u_i\rangle e_1 + \cdots + \langle Te_n, u_i\rangle e_n = \sum_j A_{i,j} e_j, \quad \text{where } T(e_j) = \sum_i A_{i,j} u_i$$

We see that T has the matrix representation of the transposition of that of T. This implies T^* have the same rank as T. Part (iii) then follows from rank-nullity theorem. ∎

Question 5

Let $\mathbb{F}$ be infinite, V be a finite dimensional $\mathbb{F}$-vector space, and $T \in \operatorname{End}(V)$. Show that

$$V \text{ is } T\text{-cyclic} \iff V \text{ has only finite number of } T\text{-invariant subspaces}$$

Proof. From now on, we denote $\mathbb{F}[x]$ by R and give V a R-module structure:

$$f \cdot w \triangleq f(T) \cdot w$$

Note that T-invariant subspaces of V then correspond to R-submodules of V.

($\implies$): Let $v \in V$ be T-cyclic, and let $m_v \in R$ be the minimal polynomial that annihilate v. Because v is T-cyclic, clearly we have a surjective R-module homomorphism

$R \twoheadrightarrow V$ defined by $f \mapsto f \cdot v$. The kernel of such module homomorphism is clearly the ideal generated by m_v. Therefore, by first isomorphism theorem, we have $V \cong R/(m_v)$. By correspondence theorem, R-submodules of $R/(m_v)$ correspond to the ideals of R that contains (m_v). Because R is a PID, this correspond to the dividing class of R that divides m_v, which clearly must be finite.

($\Longleftarrow$): By structure theorem for finitely generated modules over PID, we may write

$$V \cong R/(a_1) \oplus \cdots \oplus R/(a_n), \quad \text{where } a_1 \mid \cdots \mid a_n$$

If $V \cong R/(a_1)$, then the vector $v \in V$ correspond to 1 forms a cyclic vector. Therefore, we only have to prove $n = 1$. Assume for a contradiction that $n \geq 2$. We claim that the R-submodules

$$M_\lambda \triangleq R \cdot (1 \pmod{a_1} \times \lambda \pmod{a_2})$$

of R-module $R/(a_1) \oplus R/(a_2)$ are distinct for each $\lambda \in \mathbb{F}$. Because $\mathbb{F}$ is infinite, this would lead to a contradiction.

Let $M_\lambda = M_\mu$. It now remains to show $\lambda = \mu$. Note that $M_\lambda = M_\mu$ implies:

$$(0 \pmod{a_1} \times \lambda - \mu \pmod{a_2}) \in M_\lambda \cap M_\mu$$

This by definition of M_λ implies the existence of some $h \in R$ that makes

$$h \equiv 0 \pmod{a_1} \quad \text{and} \quad h\lambda \equiv (\lambda - \mu) \pmod{a_2}$$

Write $h = a_1 p$ for some $p \in R$. The latter now give us

$$\lambda a_1 p = (\lambda - \mu) + a_2 r, \quad \text{for some } r \in R$$

Because $a_1 \mid a_2$, we now see $a_1 \mid \lambda - \mu$, which implies $\lambda - \mu = 0$. ∎

5.2 Year 113

Question 1: standard computation

Let

$$A \triangleq \begin{bmatrix} -2 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Find the Jordan-Chevalley decomposition of A and compute e^A.

Proof. Routine computation give us

$$J = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \text{ and } P = \begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ and } A = PJP^{-1}$$

Therefore, the Jordan-Chevalley decomposition of A is

$$A = D + N \text{ where } D = P \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} P^{-1} \text{ and } N = P \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} P^{-1}$$

And

$$e^A = P \begin{bmatrix} e & 0 & 0 \\ 0 & e^{-1} & e^{-1} \\ 0 & 0 & e^{-1} \end{bmatrix} P^{-1}$$

■

Question 2: standard computation

Let V be the space of polynomial in x over $\mathbb{R}$ of degree not higher than 2. Define an inner product for V by

$$\langle f, g \rangle \triangleq \int_{-1}^{1} fg$$

Find a polynomial $k(x,t)$ such that

$$f(x) = \int_{-1}^{1} k(x,t)f(t)dt \quad \text{for all } f \in V \tag{5.1}$$

Define $T \in \text{End}(V)$ by

$$T(a_2x^2 + a_1x + a_0) \triangleq 2a_2x + a_1$$

That is, $Tf \triangleq f'$. Find the adjoint of T.

Proof. Because of the linearity of the function:

$$f(x) \mapsto g(x) \triangleq \int_{-1}^{1} k(x,t)f(t)dt$$

an endomorphism of V, for k to satisfy equation 5.1 for all $f \in V$, it only has to satisfies

$$1 = \int_{-1}^{1} kdt \text{ and } x = \int_{-1}^{1} ktdt \text{ and } x^2 = \int_{-1}^{1} kt^2dt \tag{5.2}$$

If we write k in the form

$$k = f_0(x) + f_1(x)t + f_2(x)t^2 + \cdots$$

then equation 5.2 becomes

$$\begin{aligned} 1 &= \frac{2f_0(x)}{1} + \frac{2f_2(x)}{3} + \frac{2f_4(x)}{5} + \cdots \\ x &= \frac{2f_1(x)}{3} + \frac{2f_3(x)}{5} + \frac{2f_5(x)}{7} + \cdots \\ x^2 &= \frac{2f_0(x)}{3} + \frac{2f_2(x)}{5} + \frac{2f_4(x)}{7} + \cdots \end{aligned}$$

Thus, k can be

$$k(x,t) = \Big(\frac{9}{8} + \frac{-15}{8}x^2\Big) + \Big(\frac{3x}{2}\Big)t + \Big(\frac{-15}{8} + \frac{45}{8}x^2\Big)t^2$$

Routine computation give us an orthonormal basis

$$\Big\{\sqrt{\frac{1}{2}}, \sqrt{\frac{3}{2}}x, \sqrt{\frac{45}{8}}(x^2 - \frac{1}{3})\Big\} \text{ for } V$$

With respect to this ordered basis, the matrix representation of T is

$$\begin{bmatrix} 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{15} \\ 0 & 0 & 0 \end{bmatrix}$$

Because the matrix representation of $T^\dagger$ with respect to the same basis is the conjugate transpose of that of T, we may now compute

$$T^\dagger(a_2x^2 + a_1x + a_0) = \frac{15a_1}{2}(x^2 - \frac{1}{3}) + (3a_0 + a_2)x$$

■

Question 3: equivalent definition of trace function

Let $f : M_n(\mathbb{R}) \to \mathbb{R}$ be linear. Show that if

(i) $f(AB) = f(BA)$ for all $A, B \in M_n(\mathbb{R})$. **(symmetry constraint)**

(ii) $f(I) = n$. **(initial value condition)**

then f is the trace function.

Proof. Let $E_{i,j} \in V$ be the matrix whose only nonzero entry is 1, located in the i-th row and j-th column. Because f is linear, we only have to prove

(i) If $i \neq j$, then $f(E_{i,j}) = 0$. **(f depends only on diagonal)**

(ii) $f(E_{1,1}) = \cdots = f(E_{n,n}) = 1$.

We first prove that f depends only on diagonal. Let $i \neq j$. This follows from symmetry constrain and:

$$E_{i,i}E_{i,j} = E_{i,j} \text{ and } E_{i,j}E_{i,i} = 0.$$

It remains to prove (ii). Because we have the initial value condition $f(I) = n$, we only have to show:

$$f(E_{i,i}) = f(E_{j,j}) \text{ for all } i \neq j$$

which relies on the existences of **permutation matrices**, *square matrices that has exactly one entry of 1 in each row and each column with all other entries 0.* Let $\sigma \in S_n$. Clearly, the permutation matrix:

$$C = \begin{bmatrix} \mathbf{e}_{\sigma(1)} \\ \vdots \\ \mathbf{e}_{\sigma(n)} \end{bmatrix}, \quad \text{where } \mathbf{e}_i \text{ is the } i\text{-th row of the identity matrix } I$$

when put on left side *permutes rows*:

$$CA = \begin{bmatrix} A_{\sigma(1),:} \\ \vdots \\ A_{\sigma(n),:} \end{bmatrix}$$

Because of such, its inverse C^{-1} must be

$$C^{-1} = \begin{bmatrix} \mathbf{e}_{\sigma^{-1}(1)} \\ \vdots \\ \mathbf{e}_{\sigma^{-1}(n)} \end{bmatrix} = \begin{bmatrix} (\mathbf{e}_{\sigma(1)})^t & \cdots & (\mathbf{e}_{\sigma(n)})^t \end{bmatrix}$$

which when put on right side *permutes columns*:

$$AC^{-1} = \begin{bmatrix} A_{:,\sigma(1)} & \cdots & A_{:,\sigma(n)} \end{bmatrix}$$

Let $\sigma = (i, j) \in S_n$. We now see from the symmetry constraint that:

$$f(E_{i,i}) = f(CE_{i,i}C^{-1}) = f(E_{j,i}C^{-1}) = f(E_{j,j})$$

■

Question 4: dual map rank formula: equation 5.3

Let U, V be two finite dimensional $\mathbb{F}$-vector spaces. Let $T : U \to V$ be a linear map, and let $T^\vee : U^\vee \to V^\vee$ be its dual map. Prove

$$T \text{ is injective} \iff T^\vee \text{ is surjective}$$

And

$$T \text{ is surjective} \iff T^\vee \text{ is injective}$$

Proof. Let $\{T(u_1), \dots, T(u_n)\}$ be a basis for the image of T. Extend this to a basis $\{T(u_1), \dots, T(u_n), v_1, \dots, v_m\}$ for V. Let $\{\xi_1, \dots, \xi_{n+m}\}$ be its dual basis. It is clear that $\{\xi_{n+1}, \dots, \xi_{n+m}\}$ belongs to the kernel of $T^\vee$. Observe

$$T^\vee \xi_1 u_1 = 1$$

to conclude that $\xi_i \notin \operatorname{Ker} T^\vee$ for all $1 \le i \le n$. We have shown $\{\xi_{n+1}, \dots, \xi_{n+m}\}$ is a basis for the kernel of $T^\vee$. In other words,

$$\operatorname{rank} T + \dim(\operatorname{Ker} T^\vee) = \dim V. \tag{5.3}$$

This, together with Rank-Nullity Theorem, proves both propositions. ■

Theorem 5.2.1. (Bezout's identity) If $f, g \in \mathbb{F}[x]$ makes $\gcd(f, g) = h$, then there exists $a, b \in \mathbb{F}[x]$ such that:

$$af + bg = h$$

Question 5: Bezout's identity

Let V be some finite-dimensional $\mathbb{F}$-vector space, $T \in \text{End}(V)$, and $f, g \in \mathbb{F}[x]$ two relatively prime polynomials. Prove

$$\text{Ker}(f(T)g(T)) = \text{Ker}\, f(T) \oplus \text{Ker}\, g(T).$$

Proof. We first prove that the two kernels form a direct sum. Let $v \in V$ satisfies $f(T)v = g(T)v = 0$. Because f, g are relatively prime, by Bezout's identity, there exists some $a, b \in \mathbb{F}[x]$ such that $af + bg = 1$. This implies

$$v = (af + bg)(T)v = a(T)f(T)v + b(T)g(T)v = 0$$

as desired.[1]

We now prove the equality. One side is clear:

$$\text{Ker}\, f(T) \oplus \text{Ker}\, g(T) \subseteq \text{Ker}(f(T)g(T))$$

The opposite follows from noting $v = (af)(T)v + (bg)(T)v$ and noting:

$$(af)(T)v \in \text{Ker}\, g(T) \text{ and } (bg)(T)v \in \text{Ker}\, f(T)$$

■

[1]The moral of the story here is that $(fg)(T) = f(T)g(T) = f(T) \circ g(T)$. In more precision and deeper length, it is that endomorphism $\text{End}(V)$ forms an algebra with the vector multiplication being the composition.

5.3 Year 112

Question 1: standard computation

Let

$$\mathbf{v}_1 = (1,2,0,4), \mathbf{v}_2 = (-1,1,3,-3), \mathbf{v}_3 = (0,1,-5,-2), \mathbf{v}_4 = (-1,-9,-1,-4)$$

be vectors in $\mathbb{R}^4$. Let W_1 be the subspace spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$, and let W_2 be the subspace spanned by $\mathbf{v}_3$ and $\mathbf{v}_4$. Find a basis for $W_1 \cap W_2$.

Proof. Probably $\{\mathbf{v}_3 + \mathbf{v}_4\}$. This is easy. Do it by yourself. ■

Question 2: standard computation

Let

$$A = \begin{bmatrix} 0 & 1 & -1 \\ 3 & -4 & 1 \\ 3 & -8 & 5 \end{bmatrix}$$

Find an $Q \in \mathrm{GL}_3(\mathbb{C})$ such that

$$QAQ^{-1} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 12 \\ 0 & 1 & 1 \end{bmatrix}$$

Find an invertible matrix $P \in \mathrm{GL}_3(\mathbb{C})$ such that PAP^{-1} is diagonal. *Note that my wording differs slightly with the original.*

Proof. Routine computation gives us:

$$P = \begin{bmatrix} 1 & \frac{-7}{28} & \frac{7}{28} \\ 0 & \frac{1}{28} & \frac{-1}{28} \\ -1 & \frac{36}{28} & \frac{-8}{28} \end{bmatrix} \qquad \text{and } PAP^{-1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

Let

$$B \triangleq \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 12 \\ 0 & 1 & 1 \end{bmatrix}$$

To find $Q \in \mathrm{GL}_3(\mathbb{C})$ that makes $QAQ^{-1} = B$, we first look for $M \in \mathrm{GL}_3(\mathbb{C})$ such that

$$MBM^{-1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -3 \end{bmatrix} = PAP^{-1}$$

We then see $Q \triangleq M^{-1}P$ suffices. It is then routine to compute

$$M^{-1} = \begin{bmatrix} -12 & 0 & 0 \\ -1 & 3 & 4 \\ 1 & 1 & -1 \end{bmatrix} \quad \text{and} \quad Q = M^{-1}P = \begin{bmatrix} -12 & \frac{84}{28} & \frac{-84}{28} \\ -5 & \frac{154}{28} & \frac{-42}{28} \\ 2 & \frac{-42}{28} & \frac{14}{28} \end{bmatrix}$$

■

Question 3

Define **matrix trigonometry** by

$$\sin A \triangleq \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} A^{2n+1}$$

Compute

$$\sin \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$

Show that there exist no matrix $A \in M_2(\mathbb{R})$ such that

$$\sin A = \begin{pmatrix} 1 & 2022 \\ 0 & 1 \end{pmatrix}$$

Proof. Because $\sin(PBP^{-1}) = P(\sin B)P^{-1}$ for all B and all invertible P, and because

$$\sin \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} \sin \lambda_1 & 0 \\ 0 & \sin \lambda_2 \end{pmatrix} \text{ and } \sin \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} \sin \lambda & \cos \lambda \\ 0 & \sin \lambda \end{pmatrix}$$

We may finish the proof by computing the Jordan form

$$\begin{pmatrix} 1 & 2022 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & \frac{1}{2022} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & \frac{1}{2022} \end{pmatrix}^{-1}$$

that there exists no matrix $A \in M_2(\mathbb{R})$ such that

$$\sin A = \begin{pmatrix} 1 & 2022 \\ 0 & 1 \end{pmatrix}$$

■

Question 4

Let $A = (a_{ij}) \in M_n(\mathbb{C})$, and let $\lambda_1, \dots, \lambda_n$ be roots of characteristic polynomial of A counted with multiplicity. Show that

$$A \text{ is normal} \iff \|A\|_F = \sum_{k=1}^{n} |\lambda_k|^2$$

where $\|A\|_F$ is the Frobenius norm of A.

Proof. Because the underlying field is $\mathbb{C}$, we may apply Schur's decomposition to get an upper triangular matrix D and an unitary matrix Q such that

$$A = QDQ^{-1}$$

Because A and D has the same characteristic polynomial, we know the eigenvalues of A lie on the diagonal line of D counted with multiplicity. The proof now follows from computing

$$\|A\|_F = \text{tr}(A^*A) = \text{tr}(D^*D) = \|D\|_F$$

and spectral theorem for complex finite-dimensional vector space. ■

Remark: In my opinion, the next and final question of this exam is remarkably difficult compared to the rest in this note. Someone (not me) have posted this question on MSE and received multiple great answers. As it turned out, there are multiple (distinct?) approaches. The following proof I provided is merely an expansion of one of them. The link to the post is here: https://math.stackexchange.com/q/4774695/1225113

Question 5

Let $A, B \in M_n(\mathbb{C})$. Suppose that all of the eigenvalues of A and B are positive real numbers. Prove

$$A^4 = B^4 \implies A = B$$

Proof. Because eigenvalues of A and B are all positive real, we may compute the Jordan forms of powers of Jordan blocks to see that A and B must share the same Jordan form. This implies A and B are similar, which implies the existence of some $P \in GL_n(\mathbb{C})$ such that $A = PBP^{-1}$. Clearly, to prove $A = B$, we only have to show that P and B commute. Using $A = PBP^{-1}$, we see:

$$B^4 = A^4 = PB^4P^{-1}$$

which implies P commute with B^4. To prove that P and B commute, we only have to show:

$$C(B^4) = C(B)$$

Clearly, the "$\supseteq$" part holds. To prove the "$\subseteq$" part also holds true, we only have to prove $\dim C(B^4) = \dim C(B)$, which follows from dimension formula for centralizer over algebraically closed field and computation of powers of Jordan blocks. ■

5.4 Year 110

Question 1

Let V be a finite-dimensional complex inner product space. Let $d \in \text{End}(V)$ satisfy

$$d^2 = 0$$

Let δ be the adjoint of d. Define $\Delta \in \text{End}(V)$ by

$$\Delta \triangleq d\delta + \delta d$$

Prove

(a) $\text{Ker}\, d\delta \subseteq \text{Ker}\, \delta$ and $\text{Ker}\, \delta d \subseteq \text{Ker}\, d$.

(b) $\text{Ker}\, \Delta = \text{Ker}\, d \cap \text{Ker}\, \delta$.

(c) $V = \text{Ker}\, \Delta \oplus \text{Im}\, d \oplus \text{Im}\, \delta$ is an orthogonal decomposition.

(d) $\text{Ker}\, d = \text{Ker}\, \Delta \oplus \text{Im}\, d$ is an orthogonal decomposition.

Proof. Part (a) follows from noting

$$\langle \delta v, \delta v \rangle = \langle v, d\delta v \rangle$$

and the positive definiteness of inner product. The hypothesis $d^2 = 0$ is unnecessary here. We now prove (b). The "$\supseteq$" is clear. We now prove the converse containment, which is done by unfolding the definition and premise. Let $v \in \text{Ker}\, \Delta$. Because $d\Delta = d\delta d$, we know $v \in \text{Ker}\, d\delta d$. This implies $dv \in \text{Ker}\, d\delta$, which by part (a) implies $dv \in \text{Ker}\, \delta$, and therefore by part (a) implies $v \in \text{Ker}\, d$.

We now prove (c). To prove that we have the desired orthogonal decomposition, we only have to prove

(I) $\{\text{Ker}\, \Delta, \text{Im}\, d, \text{Im}\, \delta\}$ is orthogonal.

(II) $V = \text{Ker}\, \Delta + \text{Im}\, d + \text{Im}\, \delta$

Part (I) follows from part (b) and unfolding the definitions. Noting that Δ is the sum of operators composing their adjoints, we see that Δ is self adjoint. Therefore, by spectral theorem for finite dimensional complex vector space, we have orthogonal decomposition $V = \text{Ker}\, \Delta \oplus \text{Im}\, \Delta$. Because we clearly have $\text{Im}\, \Delta \subseteq \text{Im}\, d + \text{Im}\, \delta$, part (II) follows.

We now prove (d). Note that by (c), we already know that $\operatorname{Ker}\Delta + \operatorname{Im} d$ forms an orthogonal (therefore direct) sum with dimension same as $\operatorname{Ker} d$. Therefore, we only have to show that $\operatorname{Ker}\Delta \subseteq \operatorname{Ker} d$ and $\operatorname{Im} d \subseteq \operatorname{Ker} d$. The former is clear and mentioned in (b). The latter follows from observing the Jordan form of d. ∎

Question 2: Correspondence between PD and inner product

Let $M \in \mathrm{M}_n(\mathbb{R})$ be positive definite. Let $A \in \mathrm{M}_n(\mathbb{R})$ satisfies

$$MAM^{-1} = A^t$$

Show that there exists some $P \in \mathrm{M}_n(\mathbb{R})$ such that

$$P^t MP = I \text{ and } P^{-1}AP \text{ is diagonal}$$

Proof. Recall that positive define matrix corresponds to inner product, i.e., we have the inner product $\langle \cdot, \cdot \rangle_M$ on $\mathbb{R}^n$ defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle_M \triangleq \mathbf{y}^t M \mathbf{x}$$

Denoting $P \triangleq \begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_n \end{bmatrix}$, because

$$(P^t MP)_{i,j} \triangleq (\mathbf{e}_i)^t M \mathbf{e}_j$$

We see that

$$P^t MP = I \iff \{\mathbf{e}_1, \cdots, \mathbf{e}_n\} \text{ are orthonormal with respect to } \langle \cdot, \cdot \rangle_M$$

We also have

$$P^{-1}AP \text{ is diagonal } \iff \{\mathbf{e}_1, \cdots, \mathbf{e}_n\} \text{ are eigenvectors of } A$$

The rest then follows from spectral theorem for finite dimensional real vector space and checking:

$$\langle A\mathbf{x}, \mathbf{y} \rangle_M = \mathbf{y}^t MA\mathbf{x} = \mathbf{y}^t A^t M\mathbf{x} = \langle \mathbf{x}, A\mathbf{y} \rangle_M$$

∎

Question 3

(a) Prove that invertible $M \in \operatorname{End}(\mathbb{C}^n)$ always has a square root.

(b) Let $n > 2$, and let N be a $n \times n$ matrix over some field $\mathbb{F}$ such that $N^n = 0$ and $N^{n-1} \neq 0$. Prove that there is no $n \times n$ square matrix B over $\mathbb{F}$ such that

$B^2 = N$.

Proof. For (a), we only have to show that given nilpotent $P \in \text{End}(\mathbb{C}^n)$, $I + P$ must have a square root, and then the rest of the proof will follow from decomposing M into its Jordan form. Suppose P has the nilpotency index k. That is, k is the smallest integer such that $P^k = 0$. Computing

$$(a_0 I + \cdots + a_k P^{k-1})^2 = \sum_{j=0}^{k-1} \Big(\sum_{i=0}^{j} a_i a_{j-i} \Big) P^j$$

we see that the processing of finding $a_0, \ldots, a_k \in \mathbb{R}$ to satisfy $(a_0 I + \cdots + a_k P^{k-1})^2 = I + P$ boils down to solving the following equations

$$a_0^2 = 1 \longrightarrow 2a_0 a_1 = 1 \longrightarrow 2a_0 a_2 + a_1^2 = 0 \longrightarrow 2a_0 a_3 + 2a_1 a_2 = 0 \longrightarrow \cdots$$

in order, and they clearly have the solution

$$a_0 = 1, a_1 = \frac{1}{2}, a_2 = \frac{-1}{8}, a_3 = \frac{1}{16}, a_4 = \frac{-5}{128}, a_4 = \frac{7}{256}, \ldots$$

In fact, this solution is just the coefficients of Taylor expansion of $\sqrt{1+x}$ at $x = 0$, and it has the closed form

$$a_j = \frac{\frac{1}{2}(\frac{1}{2} - 1) \cdots (\frac{1}{2} - j + 1)}{j!}$$

For (b), assume for a contradiction that $B^2 = N$, so $B^{2n} = 0$ and $B^{2n-2} \neq 0$. Various different methods show that given arbitrary nilpotent $F \in \text{End}(V)$ on finite-dimensional V, the nilpotent index of F must not be greater than $\text{Dim}(V)$. In our case, we see that $B^n = 0$, which together with $n > 2$ cause a contradiction to $B^{2n-2} \neq 0$. ■

Question 4

Let V be a vector space over some field $\mathbb{F}$, and let $u_1, \ldots, u_n \in V$ be linearly independent. Show that, for any $v_1, \ldots, v_n \in V$, the set $\{u_1 + tv_1, \ldots, u_n + tv_n\}$ is linearly independent for all but finitely many $t \in \mathbb{F}$.

Proof. Extend $\{u_1, \ldots, u_n\}$ to some basis $\{u_i : i \in I\}$ for V. Because every vector can be expressed as a finite sum of basis element, by rearranging the index, we may write for some large enough $r \in \mathbb{N}$ that

$$v_j = \sum_{i=1}^{r} b_{i,j} u_i \text{ for all } 1 \leq j \leq n$$

Clearly, $\{u_1 + tv_1, \dots, u_n + tv_n\}$ is linearly dependent only if the following determinant

$$\begin{vmatrix} 1+tb_{1,1} & tb_{1,2} & tb_{1,3} & \cdots & tb_{1,n} \\ tb_{2,1} & 1+tb_{2,2} & tb_{2,3} & \cdots & tb_{2,n} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ tb_{n,1} & tb_{n,2} & tb_{n,3} & \cdots & 1+tb_{n,n} \end{vmatrix}$$

equals to 0. This determinant is a polynomial in t over $\mathbb{F}$, which only have finite number of roots, since $\mathbb{F}$ is a field. ■

Question 5

Let P be some $n \times n$ matrix over some field $\mathbb{F}$. Show

$$\operatorname{rank}(P) + \operatorname{rank}(I - P) = n \implies P^2 = P$$

Proof. Observe

$$v = v - Pv = (I - P)v, \quad \text{for all } v \in \operatorname{Ker}(P)$$

This implies $\operatorname{Ker}(P) \subseteq \operatorname{Im}(I - P)$. It now follows from **Rank-Nullity Theorem** and

$$\operatorname{rank}(P) + \operatorname{rank}(I - P) = n$$

that

$$\operatorname{Ker}(P) = \operatorname{Im}(I - P).$$

Therefore,

$$(P - P^2)v = P(I - P)v = 0, \quad \text{for all } v \in \mathbb{F}^n \text{ as desired.}$$

■

5.5 Year 109

Question 1

Let A be a 4×4 real symmetric matrix. Suppose that 1 and 2 are eigenvalues of A and the eigenspace of 2 is 3-dimensional. Suppose $(1,-1,-1,1)^t$ is an eigenvector with eigenvalue 1.

(a) Find an orthonormal basis for the eigenspace for the eigenvalue 2 of A.

(b) Find Av, where $v = (1,0,0,0)^t$.

Proof. Let $G(2,A)$ be the eigenspace of A with eigenvalue 2. By applying spectral theorem for finite-dimensional real vector space to A, we know that

$$\mathbb{R}^4 = \text{span}\,\{(1,-1,-1,1)\} \oplus G(2,A)$$

forms an orthogonal decomposition. By computing the determinant, we know

$$\{(1,-1,-1,1),(0,1,0,0),(0,0,1,0),(0,0,0,1)\} \text{ forms a basis for } \mathbb{R}^4$$

Applying Gram-Schmidt process, we now see that

$$\left\{\frac{1}{2\sqrt{3}}(1,3,-1,1),\frac{1}{\sqrt{6}}(1,0,2,1),\frac{1}{\sqrt{2}}(-1,0,0,1)\right\} \text{ forms an orthonormal basis for } G(2,A).$$

Because

$$(1,0,0,0)\cdot\frac{1}{2}(1,-1,-1,1) = \frac{1}{2}$$

we know the orthogonal decomposition of $(1,0,0,0)$ is exactly

$$(1,0,0,0) = \frac{1}{4}(1,-1,-1,1) + \left(\frac{3}{4},\frac{1}{4},\frac{1}{4},\frac{-1}{4}\right).$$

Therefore

$$A(1,0,0,0)^t = \frac{1}{4}(1,-1,-1,1) + 2\left(\frac{3}{4},\frac{1}{4},\frac{1}{4},\frac{-1}{4}\right) = \left(\frac{7}{4},\frac{1}{4},\frac{1}{4},\frac{-1}{4}\right)$$

∎

Question 2

Let A be a real $n \times n$ matrix. Prove

$$2\operatorname{rank}(A^2) \leq \operatorname{rank}(A^3) + \operatorname{rank}(A)$$

Proof. The question is immediately solved once we decompose A into its Jordan form and recall the Jordan form of powers of Jordan blocks. ■

Question 3

Let V be a finite-dimensional real vector space, and let S, T and U be subspaces of V. Prove or disprove

(a) $\dim(S+T) = \dim(S) + \dim(T) - \dim(S \cap T)$.

(b) $\dim(S+T+U) = \dim(S) + \dim(T) + \dim(U) - \dim(S \cap T) - \dim(T \cap U) - \dim(U \cap S) + \dim(S \cap T \cap U)$.

Proof. For (a), let B be a basis for $S \cap T$, and let $B_S \subseteq S$ and $B_T \subseteq T$ satisfy

$$B_S \sqcup B, B_T \sqcup B \text{ are respectively bases for } S \text{ and } T$$

where $\sqcup$ means disjoint union. A tedious proof by contradiction shows that $B_S \cap B_T = \varnothing$ and $B_S \cup B_T$ are linearly independent. It is easy to see that $B_S \sqcup B \sqcup B_T$ spans $S + T$. We have shown $B_S \sqcup B \sqcup B_T$ forms a basis for $S + T$. This give us proposition (a).

For (b), let $V \triangleq \mathbb{R}^2, S \triangleq \operatorname{span}(1,0), T \triangleq \operatorname{span}(0,1)$ and $\mathrm{U} \triangleq \operatorname{span}(1,1)$, and we see a counter example. ■

Question 4

Let

$$M \triangleq \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$$

(a) Compute e^M.

(b) Prove $\det(e^A) = \exp(\operatorname{tr} A)$ for $A \in M_n(\mathbb{C})$.

(c) Prove or disprove

$$A \in M_n(\mathbb{C}) \text{ is nilpotent } \implies e^A - I \text{ is nilpotent}$$

Remark: the original phrasing of part (c) of this question is absolutely horrible. They should have written $\exp(A) - I$ *instead of* $\exp A - I$.

Proof. Decompose M into its Jordan form

$$M = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^{-1}.$$

We see

$$e^M = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \left(\begin{bmatrix} e & 0 \\ 0 & e \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & e \\ -e & 2e \end{bmatrix}.$$

For part (b), decompose A into its Jordan form

$$A = QJQ^{-1} = Q(S+N)Q^{-1} \text{ where } S \text{ is diagonal and } N \text{ is nilpotent.}$$

Because

$$e^{QJQ^{-1}} = \sum_{k=0}^{\infty} \frac{(QJQ^{-1})^k}{k!} = \sum_{k=0}^{\infty} \frac{QJ^kQ^{-1}}{k!} = Q\Big(\sum_{k=0}^{\infty} \frac{J^k}{k!}\Big)Q^{-1} = Qe^JQ^{-1}$$

we have

$$\det(e^A) = \det(e^{QJQ^{-1}}) = \det(Qe^JQ^{-1}) = \det(e^J)$$

Noting that e^J is upper triangular with diagonal entries e^{λ_i} where λ_i are the diagonal entries of J, we may conclude

$$\det(e^J) = e^{\sum \lambda_i} = e^{\operatorname{tr} J} = e^{\operatorname{tr}(Q^{-1}AQ)} = e^{\operatorname{tr}(A)}.$$

For part (c), let $A^m = 0$, so

$$e^A - I = \sum_{k=0}^{m-1} \frac{A^k}{k!} - I = \sum_{k=1}^{m-1} \frac{A^k}{k!}$$

We have shown $e^A - I$ is a polynomial in A with constant coefficient zero. This implies $(e^A - I)^m = 0$. ∎

Question 5

Let U, V be two finite dimensional space over the same field. Let $T : U \to V$ be a linear map, and let $T^\vee : U^\vee \to V^\vee$ be its dual map. Prove

$$T \text{ is injective} \iff T^\vee \text{ is surjective}$$

And

$$T \text{ is surjective} \iff T^\vee \text{ is injective}$$

Proof. Let $\{T(u_1), \ldots, T(u_n)\}$ be a basis for the image of T. Extend this to a basis $\{T(u_1), \ldots, T(u_n), v_1, \ldots, v_m\}$ for V. Let $\{\xi_1, \ldots, \xi_{n+m}\}$ be its dual basis. It is clear that $\{\xi_{n+1}, \ldots, \xi_{n+m}\}$ belongs to the kernel of $T^\vee$. Observe

$$T^\vee \xi_1 u_1 = 1$$

to conclude that $\xi_i \notin \operatorname{Ker} T^\vee$ for all $1 \le i \le n$. We have shown $\{\xi_{n+1}, \ldots, \xi_{n+m}\}$ is a basis for the kernel of $T^\vee$. In other words,

$$\operatorname{rank} T + \dim(\operatorname{Ker} T^\vee) = \dim V.$$

This, together with Rank-Nullity Theorem, proves both propositions. ∎

5.6 Year 108

Question 1

Find all possible Jordan forms for 8×8 real matrices $x^2(x-2)^3$ as minimal polynomial.

Proof. This question requires us to show that the Jordan form

$$J=\bigoplus_{i_1} J(\lambda_1,n_{i_1})\oplus\cdots\oplus\bigoplus_{i_m} J(\lambda_m,n_{i_m})$$

has the minimal polynomial

$$T(x)=(x-\lambda_1)^{\max_{i_1} n_{i_1}}\cdots(x-\lambda_m)^{\max_{i_m} n_{i_m}}.$$

It is clear that J is a root of T. To see that T is indeed minimal, utilize the algebraic closure of the underlying field to show that the minimal polynomial of J can not have roots other than $\lambda_1,\dots,\lambda_m$, and directly show that any polynomial that have J as a root must have power on factor $(x-\lambda_j)$ greater than $\max_{i_j} n_{i_j}$. ■

Question 2

Let V be a vector space over a field $\mathbb{F}$ of infinite elements, and let $v_1,\dots,v_n$ be vectors in V such that there exists infinite numbers of $z\in\mathbb{F}$ satisfying

$$v_0+zv_1+\cdots+z^nv_n=0$$

where $n>0$. Show that

$$v_0=v_1=\cdots=v_n=0$$

Proof. Let $\{w_\alpha:\alpha\in I\}$ be a basis for V, by rearranging the index, we may write

$$v_i=\sum_{k=1}^{r} c_{k,i}w_k \text{ for all } 1\le i\le n$$

Fix k. It remains to show $c_{k,n}=c_{k,n-1}=\cdots=c_{k,0}=0$. Clearly we have

$$c_{k,n}z^n+\cdots+c_{k,1}z+c_{k,0}=0 \text{ for infinite } z\in\mathbb{F}$$

Because $\mathbb{F}$ forms a field, we know for this to be true, we must have $c_{k,n}=c_{k,n-1}=\cdots=c_{k,0}=0$, as desired. ■

Question 3: equivalent definition of trace function

Let $f : M_n(\mathbb{R}) \to \mathbb{R}$ be linear. Show that if

(i) $f(AB) = f(BA)$ for all $A, B \in M_n(\mathbb{R})$. **(symmetry constraint)**

(ii) $f(I) = n$. **(initial value condition)**

then f is the trace function.

Proof. Let $E_{i,j} \in V$ be the matrix whose only nonzero entry is 1, located in the i-th row and j-th column. Because f is linear, we only have to prove

(i) If $i \neq j$, then $f(E_{i,j}) = 0$. **(f depends only on diagonal)**

(ii) $f(E_{1,1}) = \cdots = f(E_{n,n}) = 1$.

We first prove that f depends only on diagonal. Let $i \neq j$. This follows from symmetry constrain and:

$$E_{i,i}E_{i,j} = E_{i,j} \text{ and } E_{i,j}E_{i,i} = 0.$$

It remains to prove (ii). Because we have the initial value condition $f(I) = n$, we only have to show:

$$f(E_{i,i}) = f(E_{j,j}) \text{ for all } i \neq j$$

which relies on the existences of **permutation matrices**, *square matrices that has exactly one entry of 1 in each row and each column with all other entries 0*. Let $\sigma \in S_n$. Clearly, the permutation matrix:

$$C = \begin{bmatrix} \mathbf{e}_{\sigma(1)} \\ \vdots \\ \mathbf{e}_{\sigma(n)} \end{bmatrix}, \quad \text{where } \mathbf{e}_i \text{ is the } i\text{-th row of the identity matrix } I$$

when put on left side *permutes rows*:

$$CA = \begin{bmatrix} A_{\sigma(1),:} \\ \vdots \\ A_{\sigma(n),:} \end{bmatrix}$$

Because of such, its inverse C^{-1} must be

$$C^{-1} = \begin{bmatrix} \mathbf{e}_{\sigma^{-1}(1)} \\ \vdots \\ \mathbf{e}_{\sigma^{-1}(n)} \end{bmatrix} = \begin{bmatrix} (\mathbf{e}_{\sigma(1)})^t & \cdots & (\mathbf{e}_{\sigma(n)})^t \end{bmatrix}$$

which when put on right side *permutes columns*:

$$AC^{-1} = \begin{bmatrix} A_{;,\sigma(1)} & \cdots & A_{;,\sigma(n)} \end{bmatrix}$$

Let $\sigma = (i,j) \in S_n$. We now see from the symmetry constraint that:

$$f(E_{i,i}) = f(CE_{i,i}C^{-1}) = f(E_{j,i}C^{-1}) = f(E_{j,j})$$

■

Question 4

Let V be an n-dimensional vector space over $\mathbb{R}$ and $B : V \times V \to \mathbb{R}$ be a symmetric bilinear form.

(a) Let W be a subspace of V, and define

$$W^\perp \triangleq \{v \in V : B(v,w) = 0 \text{ for all } w \in W\}$$

Show that $\dim(W^\perp) \geq \dim(V) - \dim(W)$.

(b) Prove that $V = W \oplus W^\perp$ if and only if B is nondegenerate on W.

(c) Prove that if B is nondegenerate on V, then there exists a nonnegative integer p with $p \leq \mathrm{Dim}(V)$ and a basis $\{e_1, \dots, e_n\}$ such that

$$B(e_i, e_j) = \begin{cases} 1 & \text{if } 1 \leq i = j \leq p \\ -1 & \text{if } p+1 \leq i = j \leq n \\ 0 & \text{if } \neq j \end{cases}$$

Proof. Let $\{w_1, \dots, w_m\}$ be a basis for W. Consider the linear map $F : V \to \mathbb{R}^m$ defined by $v \mapsto (B(v,w_1), \dots, B(v,w_m))$. By Rank-Nullity Theorem, we have

$$\dim(W^\perp) = \dim(\mathrm{Ker}\, F) = \dim(V) - \mathrm{rank}(F) \geq \dim(V) - m$$

We have proved part (a). Note that $W \cap W^\perp \neq 0$ if and only if B is degenerate on W. This together with part (a) prove part (b).

Part (c) is independent of part (a) and (b). Because B is nondegenerate, we may let $\{v_1, \dots, v_n\}$ be a basis for V such that $B(v_i, v_i)$ is positive for $i \leq p$ and negative for $i > p$. Define $\widetilde{v}_1, \dots, \widetilde{v}_n$ by

$$\widetilde{v}_j \triangleq v_j - \frac{B(v_j, \widetilde{v}_{j-1})}{B(\widetilde{v}_{j-1}, \widetilde{v}_{j-1})}\widetilde{v}_{j-1} - \cdots - \frac{B(v_j, \widetilde{v}_1)}{B(\widetilde{v}_1, \widetilde{v}_1)}\widetilde{v}_1$$

It is clear that $\{\widetilde{v_1}, \dots, \widetilde{v_n}\}$ is orthogonal. It is easy to normalize them, and the result is the desired $\{e_1, \dots, e_n\}$. ■

5.7 Year 107

Question 1

Let

$$A = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 3 & 0 \\ 11 & -6 & 7 \end{bmatrix}$$

Find the lower triangular Jordan form of A. Compute e^{tA} and the general solutions of $x'(t) = Ax(t)$ where x is a 3-dimensional column vector.

Proof. Computing

$$\det(tI - A^T) = \det(tI - A) = (t-3)^3$$

and

$$\operatorname{Ker}(3I - A^T) = \operatorname{span}(\begin{bmatrix} 4 \\ -3 \\ 2 \end{bmatrix})$$

we may compute

$$A^T = \begin{bmatrix} 4 & 3 & 1 \\ -3 & -3 & -2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & \\ & 3 & 1 \\ & & 3 \end{bmatrix} \begin{bmatrix} 4 & 3 & 1 \\ -3 & -3 & -2 \\ 2 & 2 & 1 \end{bmatrix}^{-1}$$

Therefore,

$$A = \begin{bmatrix} 4 & -3 & 2 \\ 3 & -3 & 2 \\ 1 & -2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 3 & & \\ 1 & 3 & \\ & 1 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 & 2 \\ 3 & -3 & 2 \\ 1 & -2 & 1 \end{bmatrix}$$

This give us

$$
\begin{aligned}
e^{tA} &= \begin{bmatrix} 4 & -3 & 2 \\ 3 & -3 & 2 \\ 1 & -2 & 1 \end{bmatrix}^{-1} \left(\begin{bmatrix} e^{3t} & & \\ & e^{3t} & \\ & & e^{3t} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ t & 1 & 0 \\ \frac{t^2}{2} & t & 1 \end{bmatrix} \right) \begin{bmatrix} 4 & -3 & 2 \\ 3 & -3 & 2 \\ 1 & -2 & 1 \end{bmatrix} \\
&= e^{3t} \begin{bmatrix} 4 & -3 & 2 \\ 3 & -3 & 2 \\ 1 & -2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ t & 1 & 0 \\ \frac{t^2}{2} & t & 1 \end{bmatrix} \begin{bmatrix} 4 & -3 & 2 \\ 3 & -3 & 2 \\ 1 & -2 & 1 \end{bmatrix} \\
&= e^{3t} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -2 \\ -3 & 5 & -3 \end{bmatrix} \begin{bmatrix} 4 & -3 & 2 \\ 4t+3 & -3t-3 & 2t+2 \\ 2t^2+3t+1 & \frac{-3}{2}t^2 - 3t - 2 & t^2+2t+1 \end{bmatrix} \\
&= e^{3t} \begin{bmatrix} -4t+1 & 3t & -2t \\ -4t^2+2t & 3t^2+1 & -2t^2 \\ -6t^2+11t & \frac{9}{2}t^2 - 6t & -3t^2+4t+1 \end{bmatrix}
\end{aligned}
$$

The space of solutions of $x'(t) = Ax(t)$ is the column space of e^{tA}. ■

Question 2

Let V be an n-dimensional complex vector space, and let $T \in \text{End}(V)$ satisfy $T^2 = 1$.

(a) Show that T is diagonalizable.

(b) Let $S \subseteq \text{End}(V)$ be the space of endomorphisms that commute with T. Express $\dim(S)$ in term of $\text{tr}(T)$.

Proof. Clearly, T can only have eigenvalues ± 1. Recall the formula for powers of Jordan blocks. We see tha to satisfies $T^2 = 1$, each Jordan block of T must have size 1. We have shown T is indeed diagonalizable with eigenvalues ± 1. Therefore, there exists some basis under which T have the expression

$$
\begin{bmatrix} I_m & O \\ O & -I_{n-m} \end{bmatrix}
$$

Let $X \in M_n(\mathbb{C})$, and divide X into submatrices:

$$
X = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \text{ where } A \in M_m(\mathbb{C}) \text{ and } D \in M_{n-m}(\mathbb{C})
$$

We see

$$
X \in S \iff B = C = O
$$

This give us

$$\dim(S) = m^2 + (n-m)^2$$

We may now compute

$$\operatorname{tr}(T) = 2m - n \implies m = \frac{n + \operatorname{tr}(T)}{2} \implies \dim(S) = \frac{n^2 + (\operatorname{tr}(T))^2}{2}$$

■

Question 3

Let $A = (A_{ij})$ be a $2n \times 2n$ real invertible skew-symmetric matrix.

(a) Show that all eigenvalues of A are pure-imaginary.

(b) Define the Pfaffian $\operatorname{pf}(A)$ of A by

$$\operatorname{pf}(A) = \frac{1}{2^n n!} \sum_{\sigma \in S_{2n}} \operatorname{sgn}(\sigma) \prod_{k=1}^{n} A_{\sigma(2k-1),\sigma(2k)}$$

Show that for every $2n \times 2n$ real matrix B, we have

$$\operatorname{pf}(BAB^t) = \operatorname{pf}(A)\det(B)$$

(c) Suppose for some $2n \times 2n$ real orthogonal matrix O we have

$$OAO^t = \operatorname{diag}\left\{ \begin{pmatrix} 0 & m_1 \\ -m_1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & m_2 \\ -m_2 & 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 & m_n \\ -m_n & 0 \end{pmatrix} \right\}$$

where $m_i \in \mathbb{R}$ for all i, Show that $\det(A) = \operatorname{pf}(A)^2$

Proof. For part (a), one simply have to note that the matrix representation of adjoint of linear map T over $\mathbb{C}$ is just the complex conjugation of that of T. Suppose $Av = \lambda v$ for some $v \neq 0 \in \mathbb{C}^{2n}$. Because $A^* = A^t = -A$, we have

$$\lambda v \cdot v = Av \cdot v = v \cdot A^* v = -v \cdot Av = -\overline{\lambda} v \cdot v$$

This implies the real part of λ must be zero. We now prove (b). Compute

$$(BAB^t)_{ij} = \sum_{kl} B_{il} A_{lk} B_{jk}$$

Let $e_1, \dots, e_{2n}$ be the standard basis for $\mathbb{R}^{2n}$. If we define $\omega \in \bigwedge \mathbb{R}^{2n}$ by

$$\omega \triangleq \sum_{ij} (BAB^t)_{ij} e_i \wedge e_j$$

we now have

$$\omega = \sum_{ijkl} B_{il}A_{lk}B_{jk}e_i \wedge e_j = \sum_{lk} A_{lk}f_l \wedge f_k, \quad \text{where } f_p = \sum_q B_{qp}e_q$$

This together with Theorem 4.3.1 give us

$$\begin{aligned} 2^n n!\,\mathrm{pf}(A)\det(B)e_1 \wedge \cdots \wedge e_{2n} &= 2^n n!\,\mathrm{pf}(A) f_1 \wedge \cdots \wedge f_{2n} \\ &= \omega^n = 2^n n!\,\mathrm{pf}(BAB^t)e_1 \wedge \cdots \wedge e_{2n} \end{aligned}$$

The proof of part (b) now follows from noting $e_1 \wedge \cdots \wedge e_{2n} \neq 0$. We now prove (c). Again, define $\omega \in \bigwedge \mathbb{R}^{2n}$ by

$$\omega \triangleq \sum_{ij} (OAO^t)_{ij} e_i \wedge e_j$$

Because of the form of OAO^t, we see

$$\omega = 2m_1 e_1 \wedge e_2 + 2m_2 e_3 \wedge e_4 + \cdots + 2m_n e_{2n-1} \wedge e_{2n}$$

This allow us to compute

$$\omega^n = 2^n n! \Big(\prod m_i \Big) e_1 \wedge \cdots \wedge e_{2n}$$

It now follows from Equivalent Definition of Pfaffian and part (b) of this question that

$$\mathrm{pf}(A) = \mathrm{pf}(OAO^t) = m_1 \cdots m_n$$

The rest of part (c) then follows from computing $\det(A) = \det(OAO^t) = m_1^2 \cdots m_n^2$ ■

Question 4

Let $A, B \in M_n(\mathbb{C})$ be $n \times n$ complex matrices. Show that if A and B commute, then they are simultaneously triangularizable.

Proof. We have addressed simultaneous triangularization here. ■

Question 5

Show that

$$\begin{vmatrix} X_0 & X_1 & X_2 & \cdots & X_{n-1} \\ X_{n-1} & X_0 & X_1 & \cdots & X_{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ X_1 & X_2 & X_3 & \cdots & X_0 \end{vmatrix} = \prod_{j=0}^{n-1} \Big(\sum_{k=0}^{n-1} \xi^{jk} X_k \Big) \tag{5.4}$$

where ξ is a n-th primitive root of unity.

Proof. Because ξ is a n-th primitive root of unity, we may compute that the following product of $n \times n$ matrices

$$\begin{pmatrix} X_0 & X_1 & X_2 & \cdots & X_{n-1} \\ X_{n-1} & X_0 & X_1 & \cdots & X_{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ X_1 & X_2 & X_3 & \cdots & X_0 \end{pmatrix} \begin{pmatrix} \xi^0 & \xi^0 & \xi^0 & \cdots & \xi^0 \\ \xi^0 & \xi^1 & \xi^2 & \cdots & \xi^{n-1} \\ \xi^0 & \xi^2 & \xi^4 & \cdots & \xi^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \xi^0 & \xi^{n-1} & \xi^{2(n-2)} & \cdots & \xi^{(n-1)^2} \end{pmatrix}$$

is exactly

$$\begin{pmatrix} \xi^0 Y_0 & \xi^0 Y_1 & \xi^0 Y_2 & \cdots & \xi^0 Y_{n-1} \\ \xi^0 Y_0 & \xi^1 Y_1 & \xi^2 Y_2 & \cdots & \xi^{n-1} Y_{n-1} \\ \xi^0 Y_0 & \xi^2 Y_1 & \xi^4 Y_2 & \cdots & \xi^{2(n-1)} Y_{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \xi^0 Y_0 & \xi^{n-1} Y_1 & \xi^{2(n-2)} Y_2 & \cdots & \xi^{(n-1)^2} Y_{n-1} \end{pmatrix}$$

where

$$Y_j = \sum_{k=0}^{n-1} \xi^{jk} X_k \text{ for all } j$$

Noting that the right hand side of Equation 5.4 is just $\prod Y_j$ and that

$$\begin{vmatrix} \xi^0 Y_0 & \xi^0 Y_1 & \xi^0 Y_2 & \cdots & \xi^0 Y_{n-1} \\ \xi^0 Y_0 & \xi^1 Y_1 & \xi^2 Y_2 & \cdots & \xi^{n-1} Y_{n-1} \\ \xi^0 Y_0 & \xi^2 Y_1 & \xi^4 Y_2 & \cdots & \xi^{2(n-1)} Y_{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \xi^0 Y_0 & \xi^{n-1} Y_1 & \xi^{2(n-2)} Y_2 & \cdots & \xi^{(n-1)^2} Y_{n-1} \end{vmatrix} = \left(\prod_{j=0}^{n-1} Y_j \right) \begin{vmatrix} \xi^0 & \xi^0 & \xi^0 & \cdots & \xi^0 \\ \xi^0 & \xi^1 & \xi^2 & \cdots & \xi^{n-1} \\ \xi^0 & \xi^2 & \xi^4 & \cdots & \xi^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \xi^0 & \xi^{n-1} & \xi^{2(n-2)} & \cdots & \xi^{(n-1)^2} \end{vmatrix}$$

we conclude that to close out the proof with the general formula $\det(AB) = \det(A)\det(B)$, it only remains to show the determinant of the **Vandermonde matrix**

$$\begin{vmatrix} \xi^0 & \xi^0 & \xi^0 & \cdots & \xi^0 \\ \xi^0 & \xi^1 & \xi^2 & \cdots & \xi^{n-1} \\ \xi^0 & \xi^2 & \xi^4 & \cdots & \xi^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \xi^0 & \xi^{n-1} & \xi^{2(n-2)} & \cdots & \xi^{(n-1)^2} \end{vmatrix}$$

is nonzero, which we left to reader. ■

5.8 Year 106

Question 1

Let $\mathbb{F}$ be some field, and let $A \in M_n(\mathbb{F})$.

(a) Show that if k is the largest integer such that some $k \times k$ submatrix of A has a nonzero determinant, then $\operatorname{rank}(A) = k$.

(b) Suppose $A^n = 0$ and $A^{n-1} \neq 0$. Find $\max_{v \in \mathbb{F}^n} \dim(\operatorname{span}(v, Av, \dots, A^{n-1}v))$.

Proof. We first prove part (a). Because exchanging two columns or exchanging two rows do not change the rank, we may WLOG suppose the $k \times k$ submatrix is

$$\begin{pmatrix} A_{1,1} & \cdots & A_{1,k} \\ \vdots & \ddots & \vdots \\ A_{k,1} & \cdots & A_{k,k} \end{pmatrix} \tag{5.5}$$

Because the subamtrix 5.5 has nonzero determinant, we know that there exists no nonzero $(c_1, \dots, c_k) \in \mathbb{F}^k$ such that $c_1 A_{:,1} + \cdots + c_k A_{:,k} = 0$, where $A_{:,j}$ stands for the j-th column of A. We have shown $\operatorname{rank}(A) \geq k$. It remains to show $\operatorname{rank}(A) \leq k$. Assume $\operatorname{rank}(A) > k$ for a contradiction. WLOG, suppose $A_{:,1}, \dots, A_{:,k+1}$ are linearly independent. Recall that for any matrix, square or not, the columns rank equals the row rank, so there exists some $1 \leq i_1 < \cdots < i_{k+1} \leq n$ such that

$$\left\{ (A_{i_j,1}, \dots, A_{i_j,k+1}) \in \mathbb{F}^{k+1} : 1 \leq j \leq k+1 \right\} \text{ is linearly independent}$$

This implies

$$\begin{vmatrix} A_{i_1,1} & \cdots & A_{i_1,k+1} \\ \vdots & \ddots & \vdots \\ A_{i_{k+1},1} & \cdots & A_{i_{k+1},k+1} \end{vmatrix} \neq 0$$

causing a contradiction. For part (b), just note that the Jordan form always exists for nilpotent endomorphism, whether the base field is algebraically closed or not. Therefore, we may conclude the solution is

$$\max_{v \in \mathbb{F}^n} \dim(\operatorname{span}(v, Av, \dots, A^{n-1}v)) = n$$

■

Question 2

(a) Let $A = \begin{pmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{pmatrix}$. Find the general solution of linear system of ODE: $x' = Ax$.

(b) Let V be the space of all real polynomial having degree strictly less than 4 with inner product $\langle f, g\rangle \triangleq \int_{-1}^{1} f(t)g(t)dt$. Define $T \in \mathrm{End}(V)$ by $T(f) \triangleq f' + 3f$. Use the Gram-Schmidt process to replace $\beta = \{1, 1+x, x+x^2, x^2+x^3\}$ by an orthonormal basis, and find the matrix representation of the adjoint $T^\dagger$ with respect to this new basis.

Proof. The characteristic polynomial of A is $\mathrm{ch}(t) = (t-2)^2(t-1)$. Luckily, A is diagonalizable:

$$A = P\,\mathrm{diag}(1,2,2)P^{-1} \text{ where } P = \begin{pmatrix} -3 & 4 & 2 \\ 1 & 1 & 1 \\ -3 & 1 & 0 \end{pmatrix}$$

This implies

$$e^{At} = P\,\mathrm{diag}(e^t, e^{2t}, e^{2t})P^{-1} = \begin{pmatrix} -3e^t + 4e^{2t} & -6e^t - 6e^{2t} & 6e^t - 6e^{2t} \\ e^t - e^{2t} & 2e^t + 3e^{2t} & -2e^t + 2e^{2t} \\ -3e^t + 3e^{2t} & -6e^t - 6e^{2t} & 6e^t - 5e^{2t} \end{pmatrix}$$

We now do the computation for part (b). The new orthonormal basis is

$$\left\{ \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{\sqrt{2}}x, \frac{3\sqrt{5}}{2\sqrt{2}}(x^2 - \frac{1}{3}), \frac{5\sqrt{7}}{2\sqrt{2}}(x^3 - \frac{3}{5}x) \right\}$$

With respect to this basis, we have

$$[T] = \begin{pmatrix} 3 & \sqrt{3} & 0 & \sqrt{7} \\ 0 & 3 & \sqrt{15} & 0 \\ 0 & 0 & 3 & \sqrt{35} \\ 0 & 0 & 0 & 3 \end{pmatrix} \implies [T^\dagger] = \begin{pmatrix} 3 & 0 & 0 & 0 \\ \sqrt{3} & 3 & 0 & 0 \\ 0 & \sqrt{15} & 3 & 0 \\ \sqrt{7} & 0 & \sqrt{35} & 3 \end{pmatrix}$$

∎

Question 3: polar decomposition and spectral theorem

(a) Let $A \in M_n(\mathbb{R})$. Show that there exists orthonormal $Q \in M_n(\mathbb{R})$ and positive semi-definite symmetric matrix $P \in M_n(\mathbb{R})$ such that $A = QP$.[a]

(b) Let V be a finite dimensional $\mathbb{C}$-vector space, and let $T \in \text{End}(V)$. Show

$$T \text{ is normal} \iff T^\dagger = p(T) \text{ for some } p \in \mathbb{C}[x]$$

[a] I was first confused by the original phrasing of the question, so I did some googling. According to Wikipedia, the term "orthogonal matrix" and the term "orthonormal matrix" are interchangeable. They mean a real square matrix P such that $PP^t = I$. The term "positive semi-definite matrix" means a real square matrix Q such that $v^tQv \geq 0$ for all $v \in \mathbb{R}^n$

Proof. We first prove part (a). By SVD, there exists some orthonormal $U, V \in M_n(\mathbb{R})$ and diagonal $D \in M_n(\mathbb{R}_0^+)$ such that $A = UDV$. Clearly $Q \triangleq UV$ and $P \triangleq V^tDV$ suffice.

We now prove part (b). The "if" part is clear. Suppose T is normal. By spectral theorem for finite dimensional complex vector space, there exists some orthonormal basis β of V such that $[T]_\beta = \text{diag}(z_1, \dots, z_n)$ for some $z_1, \dots, z_n \in \mathbb{C}$. Because $[T^\dagger]_\beta = \text{diag}(\overline{z_1}, \dots, \overline{z_n})$. The polynomial p we are looking for only has to satisfy $p(z_i) = \overline{z_i}$ for each i. Write $\{z_1, \dots, z_n\} = \{x_1, \dots, x_k\}$ where $x_1, \dots, x_k$ are distinct. We have reduced the original question into an interpolation problem with well known solution:

$$p(x) \triangleq \prod_{i=1}^{k} \frac{\overline{x_i}(x - x_1) \cdots (x - x_{i-1})(x - x_{i+1}) \cdots (x - x_k)}{(x_i - x_1) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - z_k)}$$

■

Question 4

Let $T \in \text{End}(V)$ for finite dimensional $\mathbb{C}$-vector space V. Prove the existence and uniqueness of Jordan-Chevalley decomposition $T = S + N$. Show that there exists some polynomial $\widetilde{p} \in \mathbb{C}[x]$ such that $S = \widetilde{p}(T)$ and $\widetilde{p}(0) = 0$.

Proof. See corollary 1.2.8 ■

5.9 Year 105

Question 1

Prove the **Cayley-Hamilton Theorem**: If V is a finite dimensional vector space and $T \in \text{End}(V)$ has characteristic polynomial $p(x)$, then $p(T) = 0$.

Proof. This is a consequence of Jordan form and the fact that every field has an algebraic closure. ∎

Question 2

Let $\mathbb{F}$ be arbitrary field and $A \in M_{m\times n}(\mathbb{F})$. Show that:

(i) The row ranks of A equals the column ranks of A.

(ii) If A have rank r, then AA^t also have rank r.

Proof. Suppose $\text{rank}(A) = r$. Let $\alpha \triangleq \{e_1, \dots, e_n\}$ and $\beta \triangleq \{e'_1, \dots, e'_m\}$ respectively be basis for $\mathbb{F}^n$ and $\mathbb{F}^m$ such that $\{A(e_1) = e'_1, \dots, A(e_r) = e'_r\}$ form a basis for image of A, and let γ and δ respectively be the standard basis for $\mathbb{F}^n$ and $\mathbb{F}^m$. Let $\alpha^\vee \triangleq \{\phi_1, \dots, \phi_n\}, \beta^\vee \triangleq \{\psi_1, \dots, \psi_m\}, \gamma^\vee$, and $\delta^\vee$ be the dual bases. Let $A^\vee : (\mathbb{F}^m)^* \longrightarrow (\mathbb{F}^n)^*$ be the dual map of A. After setting these definitions, to get full credits of the first problem, one only need to show

$$[A^\vee]_{\delta^\vee}^{\gamma^\vee} = A^t$$

and show

$$A^\vee(\psi_i) = \begin{cases} \varphi_i & \text{if } i \leq r \\ 0 & \text{if } i > r \end{cases} \tag{5.6}$$

They are all routine to check. The second problem seems to be wrong. Consider the field $\mathbb{F} = \mathbb{Z}_2$, and

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

∎

Question 3

Let $\mathbb{F}$ be arbitrary field and let $\mathcal{C}$ be a possibly infinite collection of $n \times n$ $\mathbb{F}$-matrices that are all diagonalizable and commute. Prove that there exists some $P \in GL_n(\mathbb{F})$

such that PA_iP^{-1} are diagonal for all $i \in I$.

Proof. Because $\text{End}(\mathbb{F}^n)$ is finite dimensional, there exists some finite subset of $\mathcal{C}$ whose spans contains all $\mathcal{C}$. Clearly, if there exists a basis that contains only eigenvectors of the elements this finite subset, then every linear operator in $\mathcal{C}$ with respect to this basis is diagonal, since each of them is a finite linear combination of the elements of the finite subset. Because of such, we may suppose $\mathcal{C}$ is finite. Write $\mathcal{C} \triangleq \{A_1, \dots, A_m\}$. Let Λ_i be the set of eigenvalues of A_i. It remains to show

$$\mathbb{F}^n = \bigoplus_{\lambda_i \in \Lambda_i} (E(\lambda_1, A_1) \cap \cdots \cap E(\lambda_m, A_m)) \tag{5.7}$$

We shall show equation 5.7 by induction on m. The base case is when $m = 2$. Let $\lambda \in \Lambda_1$. Clearly, for each $v \in E(\lambda, A_1)$, we have

$$(A_1 - \lambda I)(A_2 v) = A_2(A_1 - \lambda I)v = 0 \tag{5.8}$$

We have shown eigenspaces with respect to A_1 are all stable under A_2. Because restriction of diagonalizable linear operator is still diagonalizable, for each $\lambda \in \Lambda_1$, we have

$$E(\lambda, A_1) = \bigoplus_{\mu \in \Lambda_2} E(\mu, A_2|_{E(\lambda, A_1)}) = \bigoplus_{\mu \in \Lambda_2} E(\lambda, A_1) \cap E(\mu, A_2)$$

where the second equality hold trivially. The base case $m = 2$ now follows from $\mathbb{F}^n = \bigoplus_{\lambda \in \Lambda_1} E(\lambda, A_1)$. We now prove the inductive case. Suppose

$$\mathbb{F}^n = \bigoplus_{\lambda_i \in \Lambda_i} (E(\lambda_1, A_1) \cap \cdots \cap E(\lambda_{m-1}, A_{m-1}))$$

Fix $\lambda_i \in \Lambda_i$. Again for the same reason, similar to equation 5.8, for each $i \leq m - 1$, eigenspace $E(\lambda_i, A_i)$ is stable under A_m, so clearly $V \triangleq E(\lambda_1, A_1) \cap \cdots \cap E(\lambda_{m-1}, A_{m-1})$ is stable under A_m. Again because restriction of diagonalizable linear operator is still diagonalizable, we have

$$V = \bigoplus_{\lambda \in \Lambda_m} E(\lambda, A_m|_V) = \bigoplus_{\lambda \in \Lambda_m} E(\lambda_1, A_1) \cap \cdots \cap E(\lambda_{m-1}, A_{m-1}) \cap E(\lambda, A_m)$$

where the second equality hold by definition of V as one can check. We have shown equation 5.7. ∎

Question 4

Prove the **Spectral Theorem for finite dimensional complex vector space**.

Proof. This has been addressed in the section on spectral theorem. Long story short, you consider Jordan form, use Gram-Schmidt process to get Schur's form, and use the hypothesis of the linear operator being normal to deduce that its Schur's form is in fact diagonal, via direct computation. ■

Question 5

Let V be a finite-dimensional complex vector space, and let $T \in \operatorname{End}(V)$ has characteristic polynomial p. Suppose $p = q_1 q_2$ for some q_1 and $q_2 \in \mathbb{C}[x]$ that have no common roots in $\mathbb{C}$. Show that there exists subspace W_1 and $W_2 \subseteq V$ that satisfies

(i) $V = W_1 \oplus W_2$.

(ii) W_1 and W_2 are both stable under T.

(iii) the characteristic polynomials of $T|_{W_1}$ and $T|_{W_2}$ are respectively q_1 and q_2.

Proof. This is a consequence of primary decomposition part of structure theorem for finitely generated modules over PID. ■

5.10 Year 104

Question 1

(i) For each $x \in \mathbb{R}$, let V_x be the subspace of $\mathbb{R}^4$ spanned by

$$(x, 1, 1, 1), (1, x, 1, 1), (1, 1, x, 1), (1, 1, 1, x)$$

Determine all $x \in \mathbb{R}$ such that $\dim(V_x) \leq 3$.

(ii) Find the dimension and a basis for the space of $\mathbb{R}$-linear maps from $\mathbb{R}^5$ to $\mathbb{R}^3$ whose kernel contains $(0, 2, -3, 0, 1)$.

Proof. The first problem boils down to factorizing the determinant:

$$\begin{vmatrix} x & 1 & 1 & 1 \\ 1 & x & 1 & 1 \\ 1 & 1 & x & 1 \\ 1 & 1 & 1 & x \end{vmatrix} \tag{5.9}$$

into linear terms. Even though one may just perform direct computation and "guesses work" to find the solutions, we offer a different approach: Because Matrix 5.9 equals to $(x-1)I + J$ where all entries of $J \in M_4(\mathbb{R})$ are 1, by Theorem 5.10.1, the determinant of Matrix 5.9 is $(x-1)^3(x-1+4)$. This implies the solutions of the first problem is $\{1, -3\}$.

In the second problem, if we identify $\text{Hom}_{\mathbb{R}}(\mathbb{R}^5, \mathbb{R}^3)$ with $M_{3\times 5}(\mathbb{R})$ the natural way, then the space in the question consists precisely of those matrix whose every row is orthogonal to $(0, 2, -3, 0, 1)$. Clearly, the subspace of $\mathbb{R}^5$ consisting of vectors orthogonal to $(0, 2, -3, 0, 1)$ is 4-dimensional, and says, if $\{\mathbf{v}_1, \dots, \mathbf{v}_4\}$ is a basis, then the space in the question is exactly 12-dimensional with basis $\{E_{i,j} : 1 \leq i \leq 3, 1 \leq j \leq 4\}$ where the non i-th row of $E_{i,j}$ are all 0 and the i-th row of $E_{i,j}$ is $\mathbf{v}_j$. We have reduced the problem into finding $\{\mathbf{v}_1, \dots, \mathbf{v}_4\}$. Clearly, $\mathbf{v}_1 \triangleq (1, 0, 0, 0, 0), \mathbf{v}_2 \triangleq (0, 1, 1, 0, 1), \mathbf{v}_3 \triangleq (0, 1, 0, 0, -2), \mathbf{v}_4 \triangleq (0, 0, 0, 1, 0)$ suffices. ∎

Theorem 5.10.1. (Determinant of special matrices) Let $\mathbb{F}$ be some field, $\alpha, \beta \in \mathbb{F}$, and every entry of $J \in M_n(\mathbb{F})$ be 1. The matrix $\alpha I + \beta J$ is $\mathbb{F}$-diagonalizable with eigenvalues:

$$\overbrace{\alpha, \alpha, \dots, \alpha}^{n-1 \text{ copies}}, \quad \text{and} \quad \alpha + n\beta \tag{5.10}$$

Therefore,

$$\det(\alpha I + \beta J) = \alpha^{n-1}(\alpha + n\beta) \tag{5.11}$$

Proof. Noting $\text{rank}_{\mathbb{F}}(J) = 1$, we see that J is $\mathbb{F}$-diagonalizable with eigenvalues:[2]

$$\overbrace{0, 0, \ldots, 0}^{n-1 \text{ copies}}, \quad \text{and} \quad n$$

Let $\mathbf{v} \in \mathbb{F}^n$. Clearly, $J\mathbf{v} = \mathbf{0} \implies (\alpha I + \beta J)\mathbf{v} = \alpha\mathbf{v}$. Also, clearly, $J\mathbf{v} = n\mathbf{v} \implies (\alpha I + \beta J)\mathbf{v} = (\alpha + n\beta)\mathbf{v}$. This proves Equation 5.10. Now to see Equatoin 5.11 holds, just consider its Jordan form. ∎

Question 2

Let

$$A \triangleq \begin{bmatrix} -1 & 4 & -2 \\ -2 & 5 & -2 \\ -1 & 2 & 0 \end{bmatrix}$$

(i) Compute the characteristic polynomial of A.

(ii) If $f(x) \triangleq (x-3)^2 + 5$, find the eigenvalue of $f(A)$.

(iii) Find an orthogonal matrix $P \in M_3(\mathbb{R})$ such that $P^{-1}AP$ is diagonal.

Proof. The characteristic polynomial of A is $(x-2)(x-1)^2$. By spectral theorem of polynomial, the set of eigenvalues of $f(A)$ is then exactly $\{6, 9\}$. The third question is clearly wrong: There is no such P. By spectral theorem for finite dimensional real vector space, for such P to exist, A must be symmetric. ∎

Question 3

Let $A, B \in \text{GL}_n(\mathbb{R})$. Show that

(i) If $ABA^{-1}B^{-1} = cI$, then $c = \pm 1$.

(ii) If $AB - BA = cI$, then $c = 0$.

Proof. Note that $ABA^{-1}B^{-1} = cI$ implies $AB = cBA$, which implies $\det(AB) = c^n \det(BA)$, which implies $c^n = 1$, which implies $c \in \{-1, 1\}$. Note that $AB - BA = cI$ implies $AB = cI + BA$, which implies $\text{trace}(AB) = nc + \text{trace}(BA)$, which implies $nc = 0$, which implies $c = 0$. ∎

---

[2]Note that here, by $n \in \mathbb{F}$, we mean $\overbrace{1 + \cdots + 1}^{n \text{ copies}} \in \mathbb{F}$

Question 4

Given $A \in M_n(\mathbb{R})$, show

$$A^3 = A \implies \text{rank}(A) = \text{tr}(A^2)$$

Proof. $A^3 = A$ implies that $x^3 - x = x(x-1)(x+1)$ is a polynomial that send A to 0. By long division, minimal polynomial of A must divides $x(x-1)(x+1)$. This implies that the set of eigenvalues of A is contained by $\{-1, 0, 1\}$. It then follows from considering the Jordan form of A that $\text{rank}(A) = \text{tr}(A^2)$. ■

Question 5

Given $A \in M_n(\mathbb{R})$, show

$$\text{rank}(A) + \text{rank}(I - A) = n \implies A^2 = A$$

Proof. Suppose A has distinct eigenvalues $\lambda_1 = 0, \lambda_2 = 1, \lambda_3, \dots, \lambda_k \in \mathbb{C}$ with algebraic dimension d_i (So $d_i = 0$ if λ_i is not an eigenvalue), and suppose the Jordan form of A is

$$\bigoplus_{i=1}^{k} J(\lambda_i, r_1) \oplus \cdots \oplus J(\lambda_i, r_{m_i})$$

Compute

$$\text{rank}(A) = n - m_1 \text{ and } \text{rank}(I - A) = n - m_2$$

This by premise give us

$$m_1 + m_2 = n$$

which implies that the $d_2 = \cdots = d_k = 0$ and all Jordan blocks of A have size 1. This implies that the minimal polynomial of A is either $x, x-1$, or $x(x-1)$. Either way, this implies $A^2 = A$. ■

Question 6

Consider $A, B \in M_n(\mathbb{Q})$ with $A^n = 0$ but $A^{n-1} \neq 0$. Show

$$AB = BA \implies B = F(A) \text{ for some } F \in \mathbb{Q}[x] \text{ such that } \deg(F) < n$$

Proof. Looking at the Jordan form of A, we know there exists some $\mathbf{e} \in \mathbb{Q}^n$ such that $\{\mathbf{e}, A\mathbf{e}, \dots, A^{n-1}\mathbf{e}\}$ forms a basis for $\mathbb{Q}^n$. Therefore, there exists some $c_0, \dots, c_{n-1} \in \mathbb{Q}$

such that if we define $F(x) \triangleq c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$, then $F(A)$ and B have the same action on $\mathbf{e}$. To see such F suffices, we only have to show B and $F(A)$ moreover agree on rest of the $\left\{\mathbf{e}, A\mathbf{e}, \ldots, A^{n-1}\mathbf{e}\right\}$, which is a consequence of the commutation:

$$BA^i\mathbf{e} = A^iB\mathbf{e} = A^iF(A)\mathbf{e} = F(A)A^i\mathbf{e}$$

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